Convex Hulls

Sujoy Bhore

Indian Institute of Technology Bombay

Mixing Ratios

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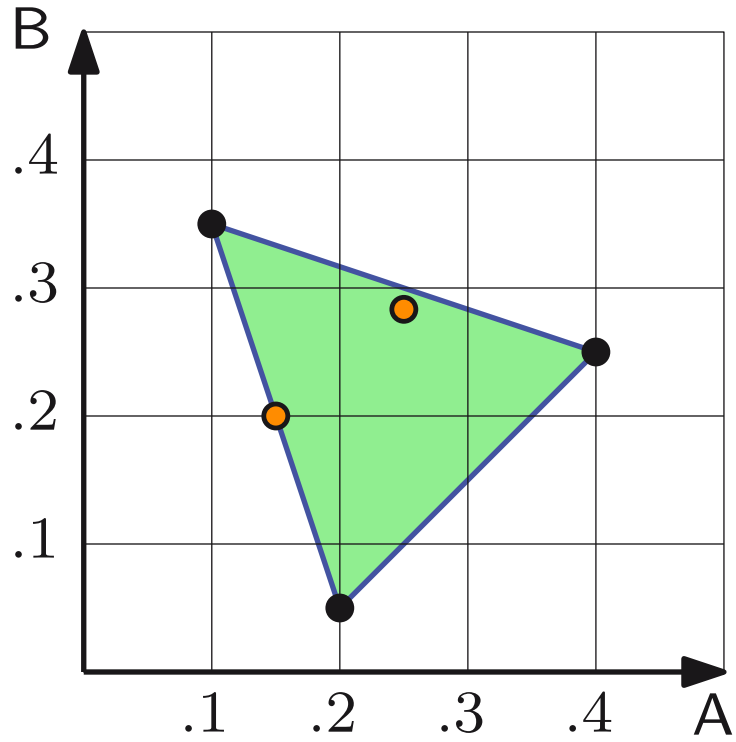

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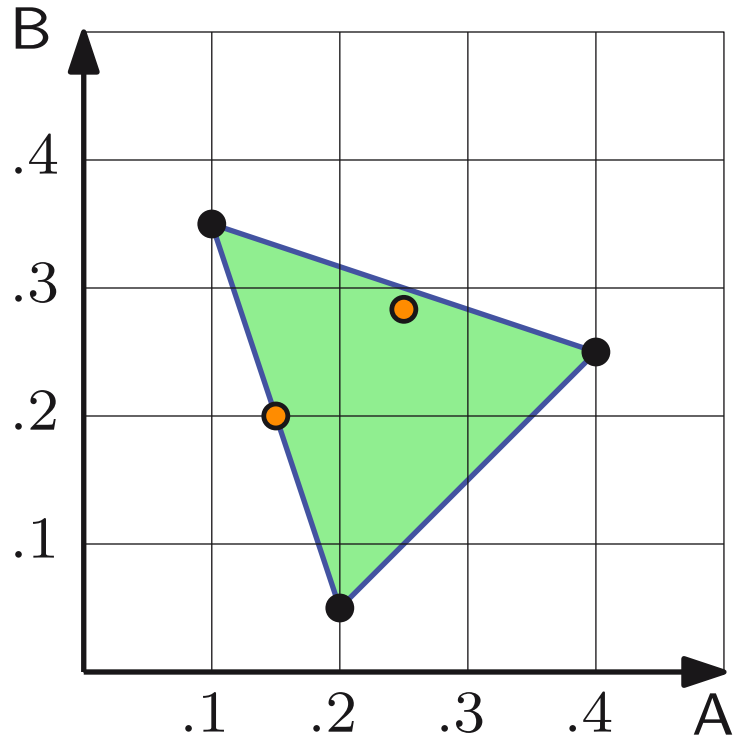

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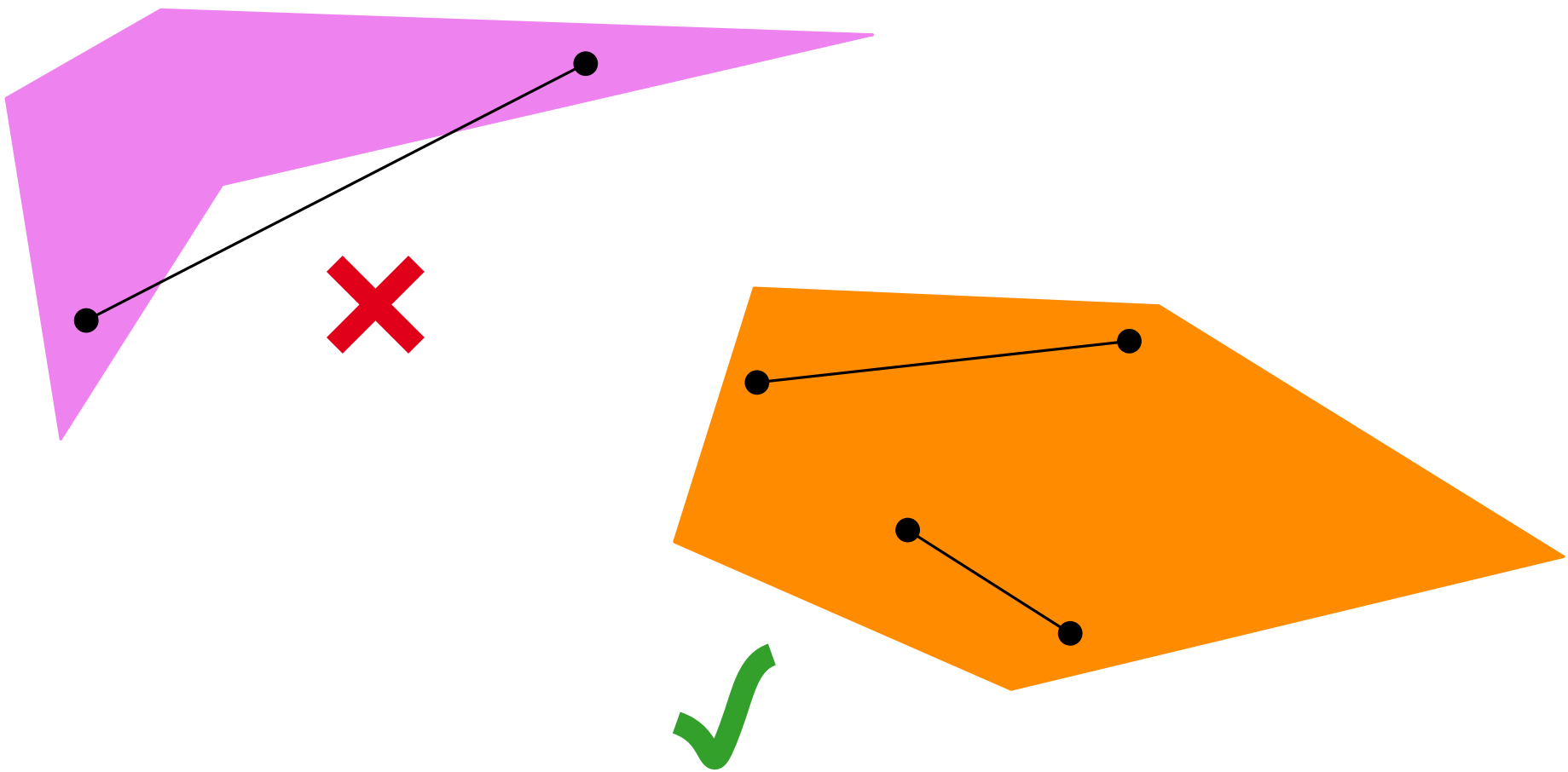
Observation. Given a set $S \subseteq \mathbb{R}^2$ mixtures, it is possible to make another mixture $q \in \mathbb{R}^2$ using $S \iff q \in$ ConvexHull $CH(S)$.
$q = \sum_i \lambda_i s_i$ with $\sum_i \lambda_i = 1$.

Definition of Convex Hull

Definition. A region $S \subseteq \mathbb{R}^2$ is called **convex**, if for any two points $p, q \in S$ the line segment $\overline{pq} \in S$. The **convex hull** $CH(S)$ of S is the smallest convex region containing S.



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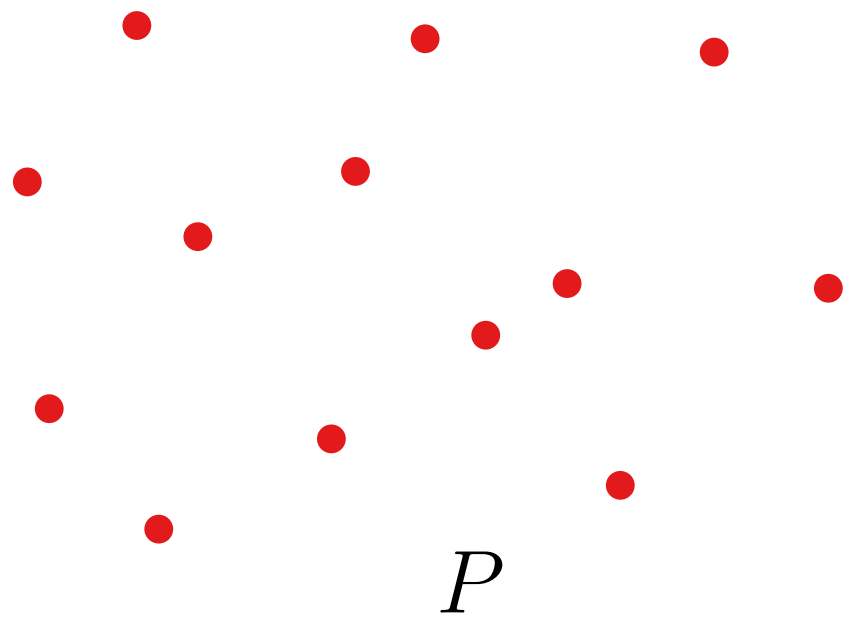

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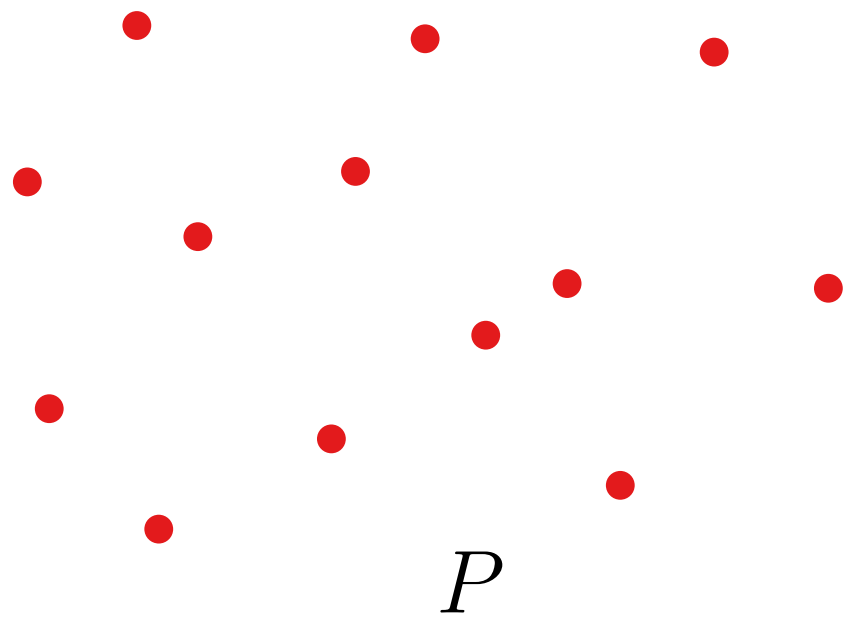

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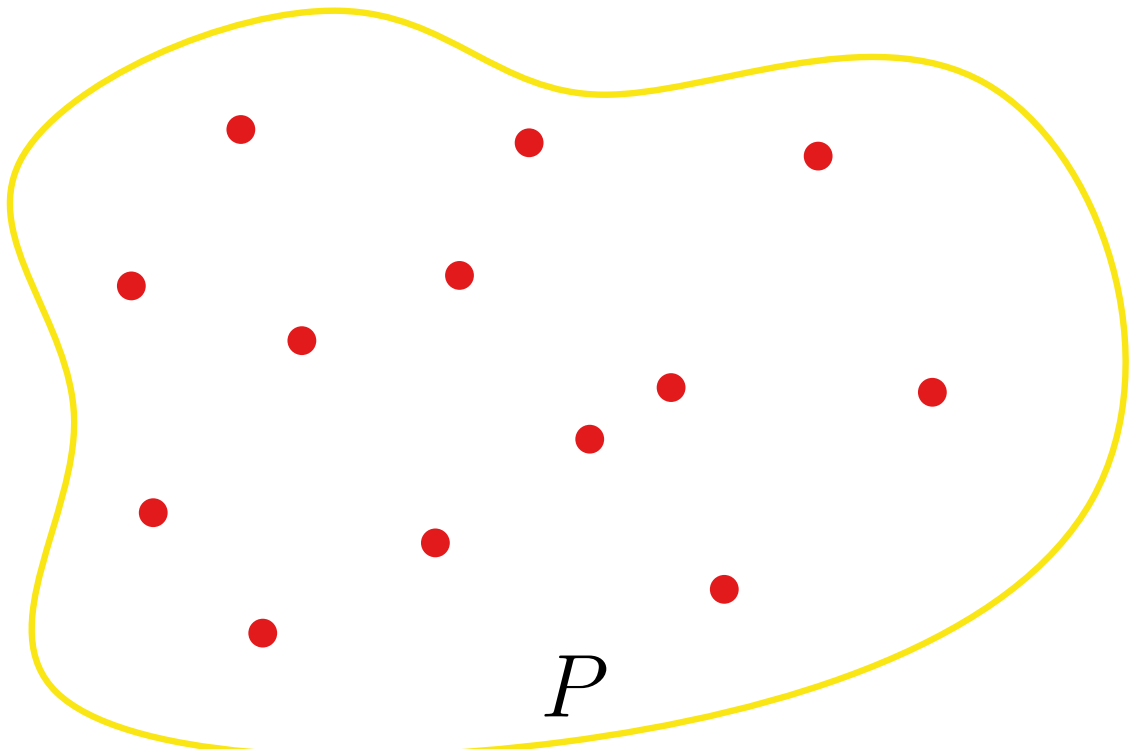

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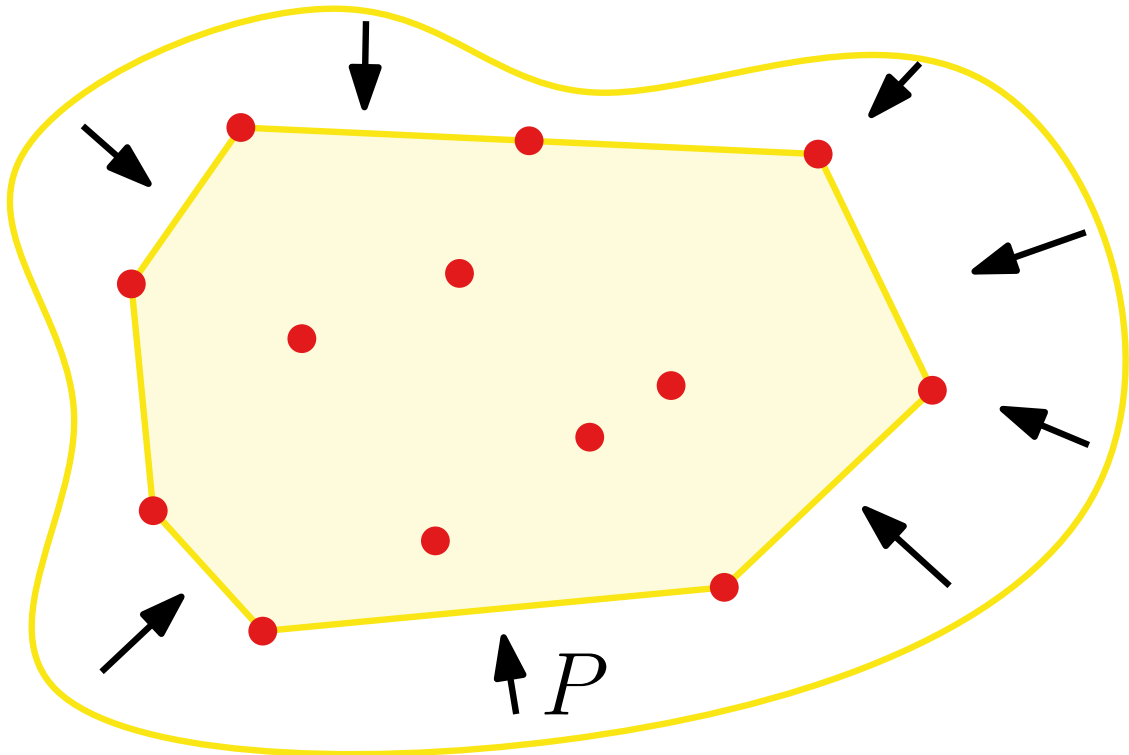

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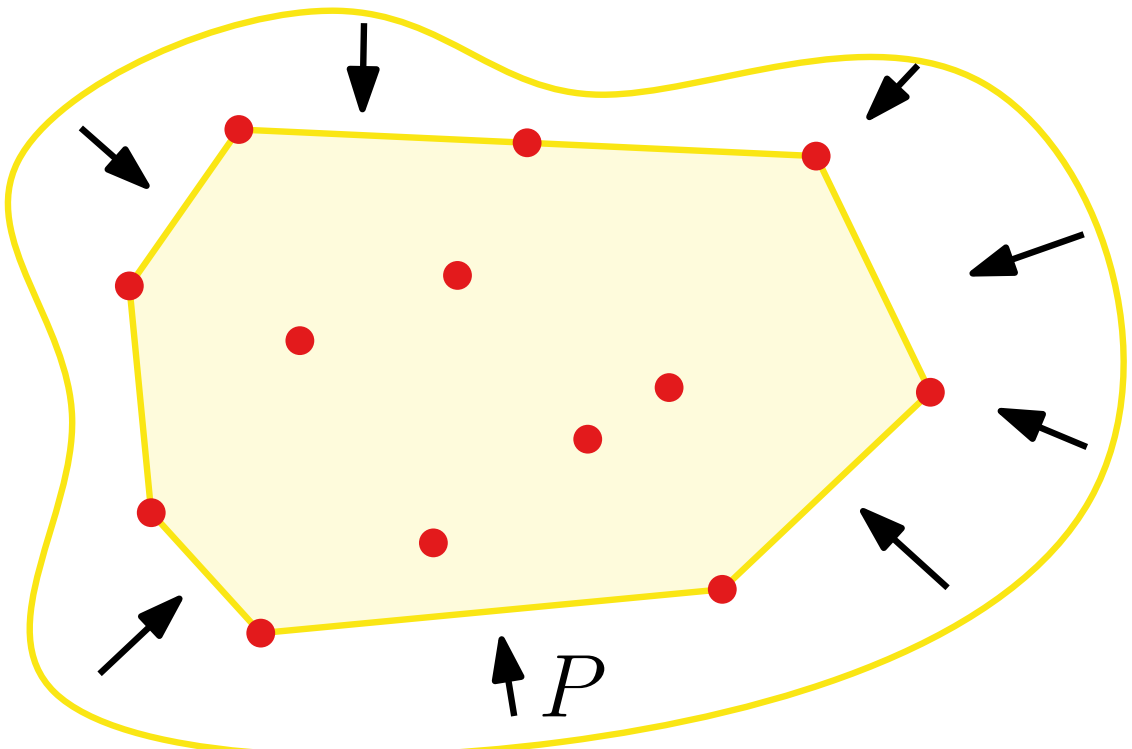

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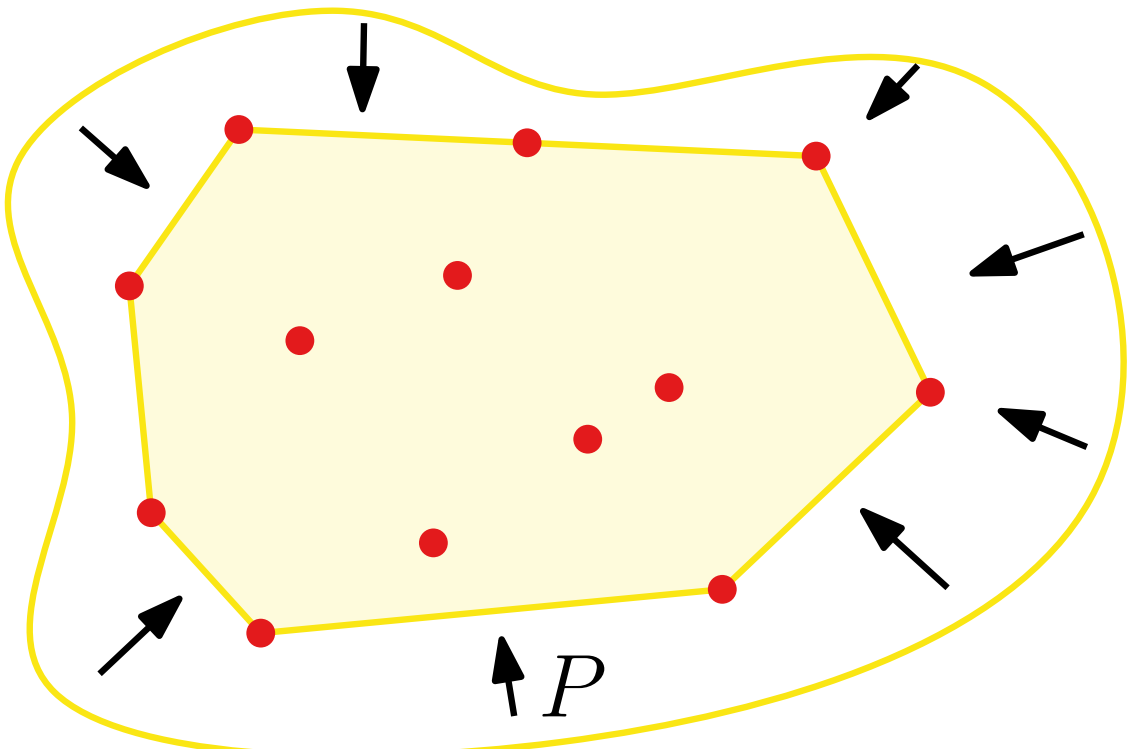

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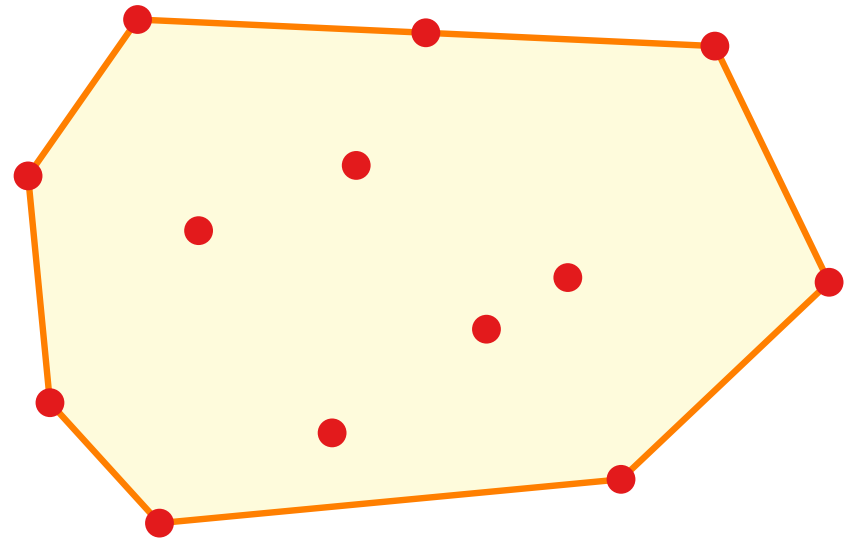
Now, some Mathematics -

- define $CH(S) = \bigcap_{C \supseteq S:\ C \text{ convex}} C$
- does not help either.

Algorithmic Approach

Lemma. For a set of points $P \subseteq \mathbb{R}^2$, $CH(P)$ is a convex polygon that contains P and whose vertices are a subset of P.



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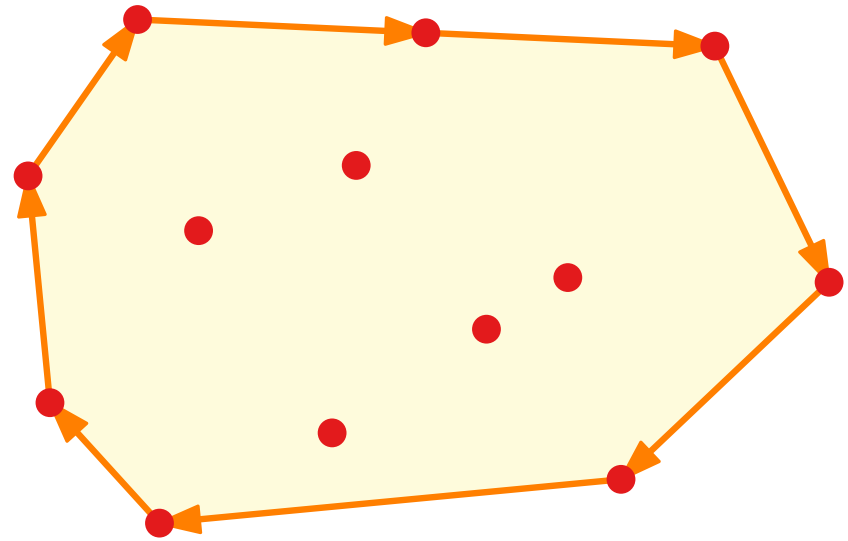

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Output: List of vertices of $CH(P)$ in clockwise order

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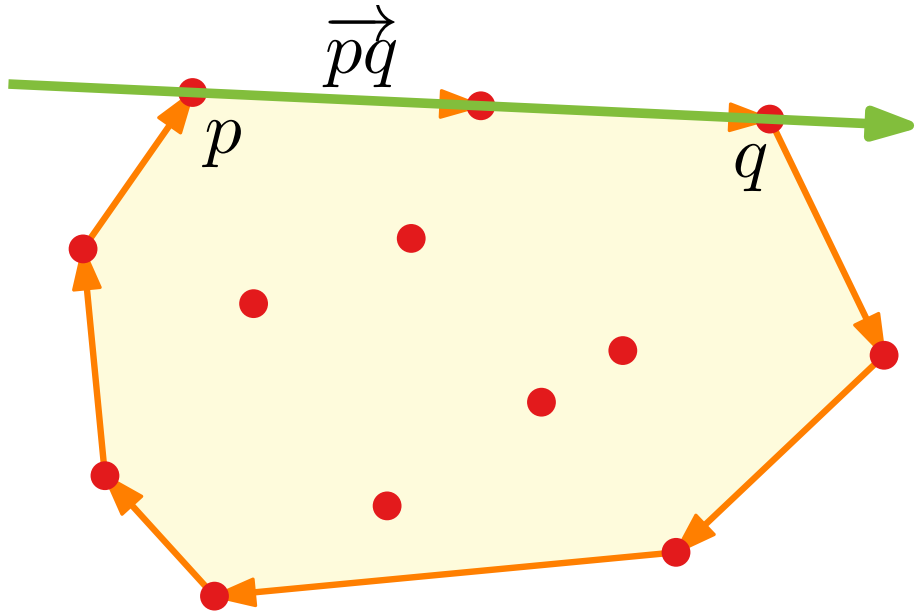

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Observation. (p, q) is an edge of $CH(P) \Leftrightarrow$ each point $r \in P \setminus \{p, q\}$

- strictly right of the oriented line $\overrightarrow{pq}$ or
- on the line segment $\overline{pq}$

Algorithm - 1

ConvexHull(P)

$E \leftarrow \emptyset$

foreach $(p, q) \in P \times P$ with $p \neq q$ **do**

  valid $\leftarrow$ *true*

  foreach $r \in P$ **do**

    if not (r strictly right of $\overrightarrow{pq}$ **or** $r \in \overline{pq}$) **then**

      valid $\leftarrow$ *false*

  if *valid* **then**

    $E \leftarrow E \cup \{(p, q)\}$

construct sorted vertex list L of $CH(P)$ from E

return L

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Test in $O(1)$ time with

$$\begin{vmatrix} x_r & y_r & 1 \\ x_p & y_p & 1 \\ x_q & y_q & 1 \end{vmatrix} < 0$$

$\rightarrow$ think (exercise-1)

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Running Time Analysis

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Lemma. The convex hull of n points in $\mathbb{R}^2$ can be computed in $O(n^3)$ time.

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Can we do better?

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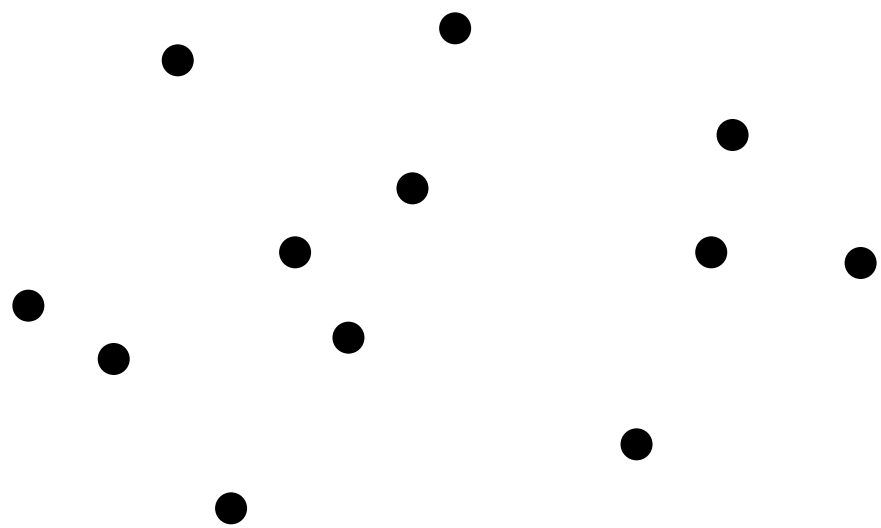

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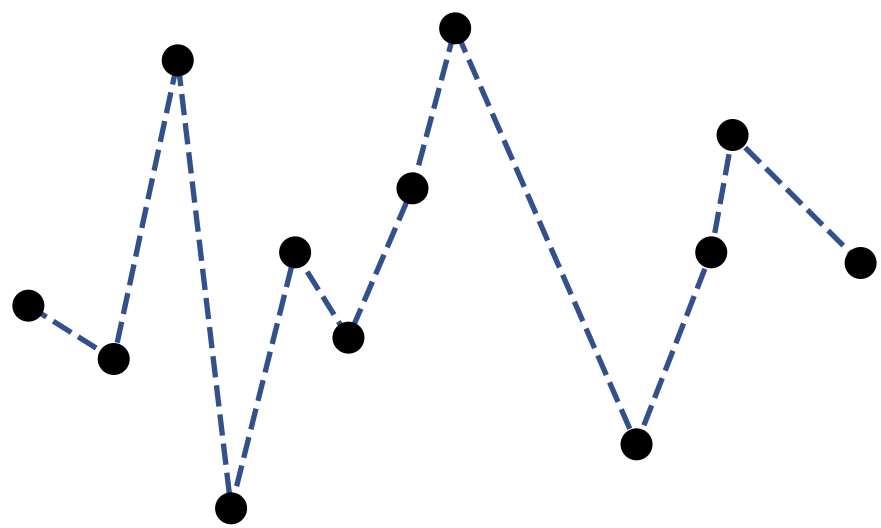

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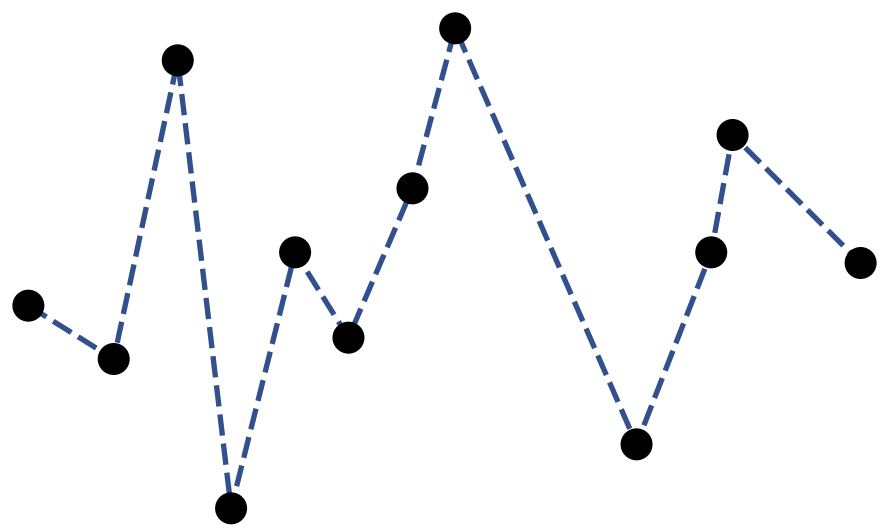

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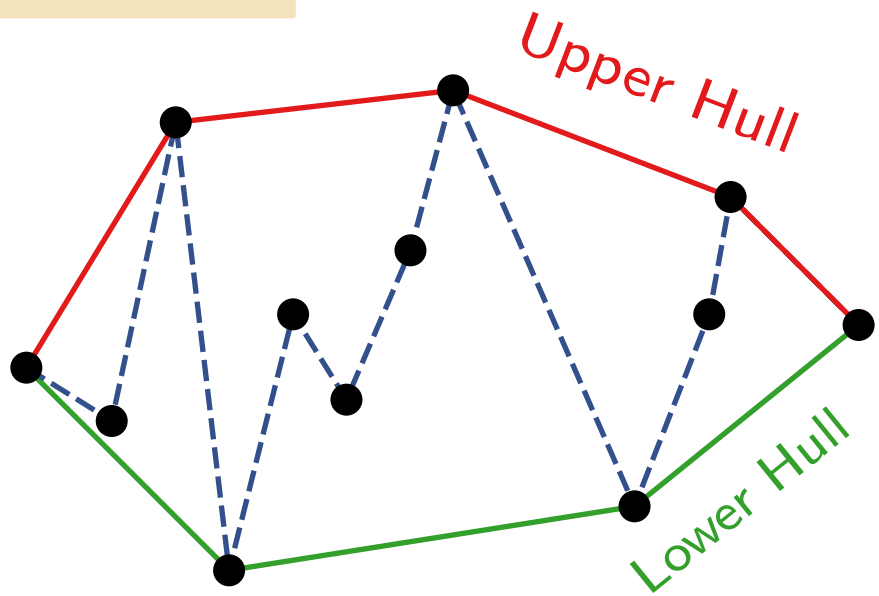

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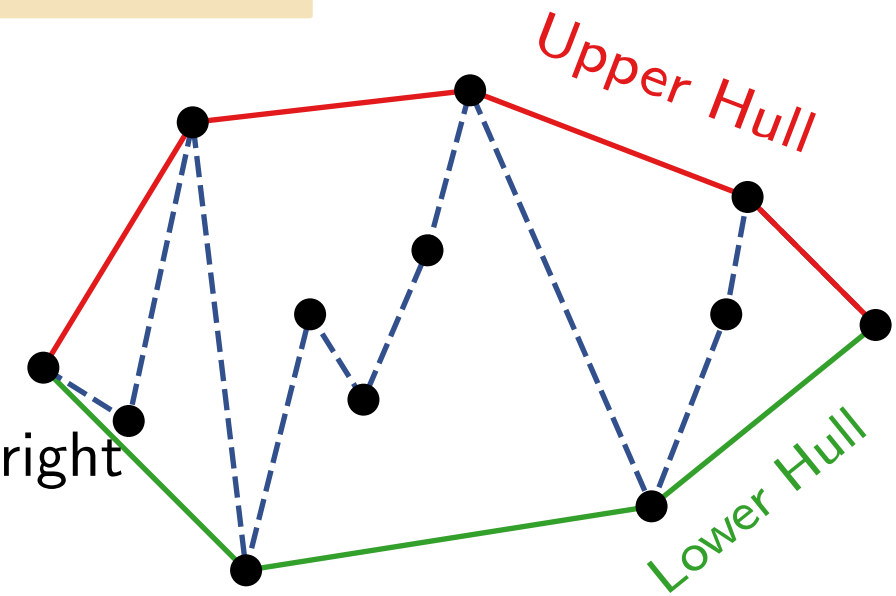

UpperConvexHull(P)
  $\langle p_1, p_2, \ldots, p_n \rangle \leftarrow$ sort P from left to right
  $L \leftarrow \langle p_1, p_2 \rangle$
  for $i \leftarrow 3$ **to** n **do**
    L.append(p_i)
    while $|L| > 2$ **and** last 3 points in L do not form right turn **do**
      remove the second-to-last point in L
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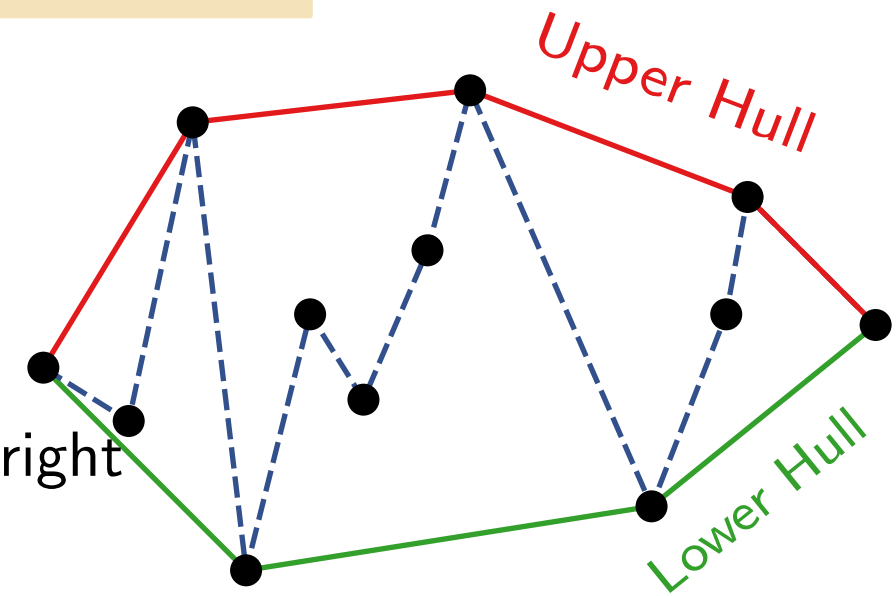

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lower hull handled similarly!

Analysis

Running Time -

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Analysis

Running Time -

UpperConvexHull(P)

- $\langle p_1, p_2, \ldots, p_n \rangle \leftarrow$ sort P from left to right — $O(n \log n)$
- $L \leftarrow \langle p_1, p_2 \rangle$
- **for** $i \leftarrow 3$ **to** n **do** — $n - 2$
  - L.append(p_i)
  - **while** $|L| > 2$ **and** last 3 points in L do not form right turn **do** — $O(n)$
    - remove the second-to-last point in L
- **return** L

Analysis

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$\langle p_1, p_2, \ldots, p_n \rangle \leftarrow$ sort P from left to right $\quad O(n \log n)$

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Amortized Analysis

- Each point is inserted into L exactly once
- A point in L is removed at most once from L
- $\Rightarrow$ Running time of the **for** loop including the **while** loop is $O(n)$

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Theorem. The convex hull of n points in the plane can be computed in $O(n \log n)$ time. $\rightarrow$ *Graham's Scan*.

Let's do a Gift Wrapping approach ...

Idea: Begin with a point p_1 of $CH(P)$, then find the next edge of $CH(P)$ in clockwise order.

GiftWrapping(P)

$p_1 = (x_1, y_1) \leftarrow$ rightmost point in P; $p_0 \leftarrow (x_1, \infty)$; $j \leftarrow 1$
while true **do**
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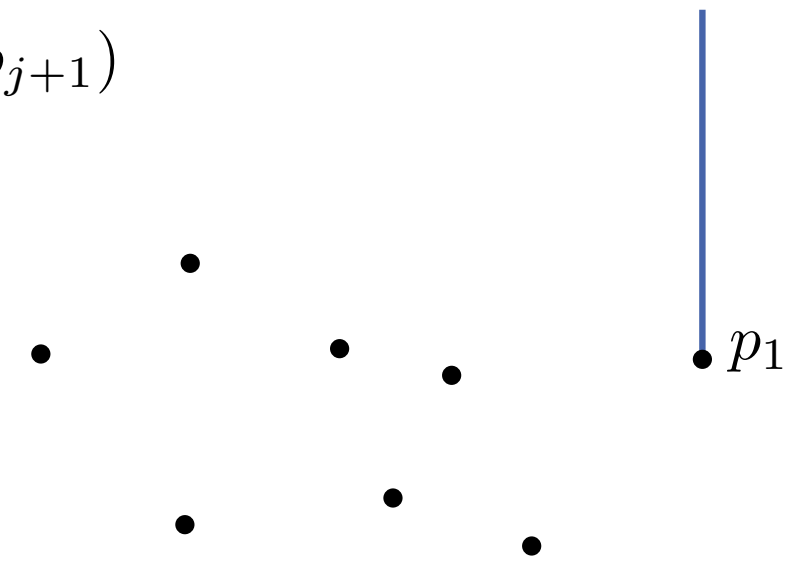

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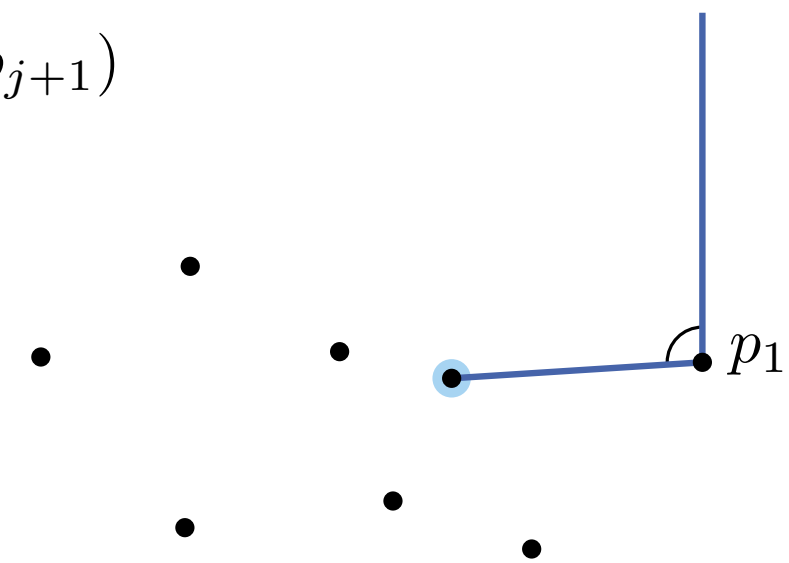

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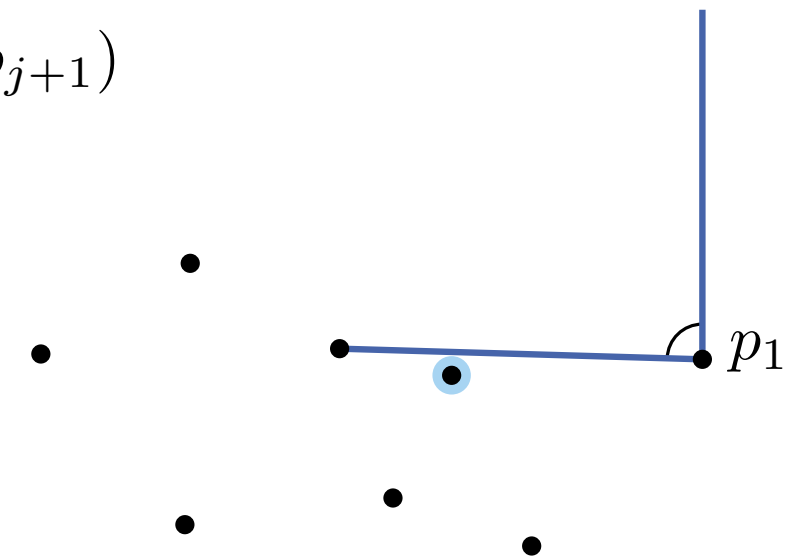

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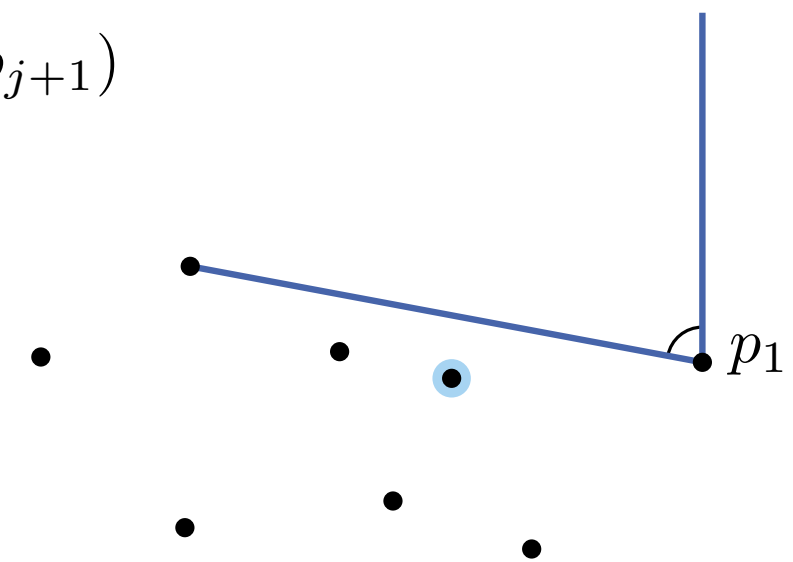

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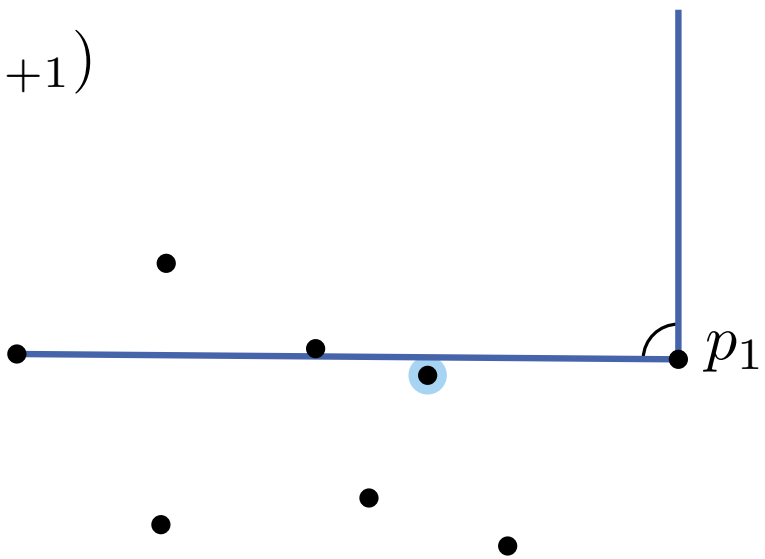

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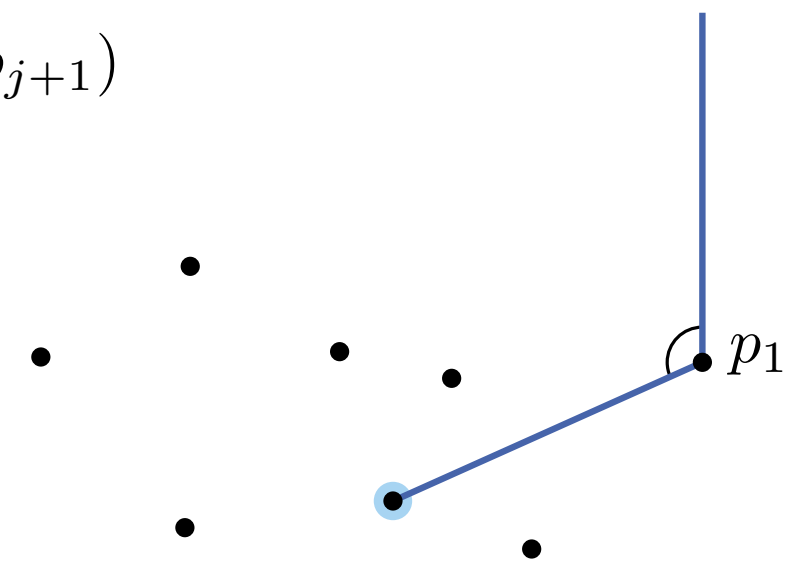

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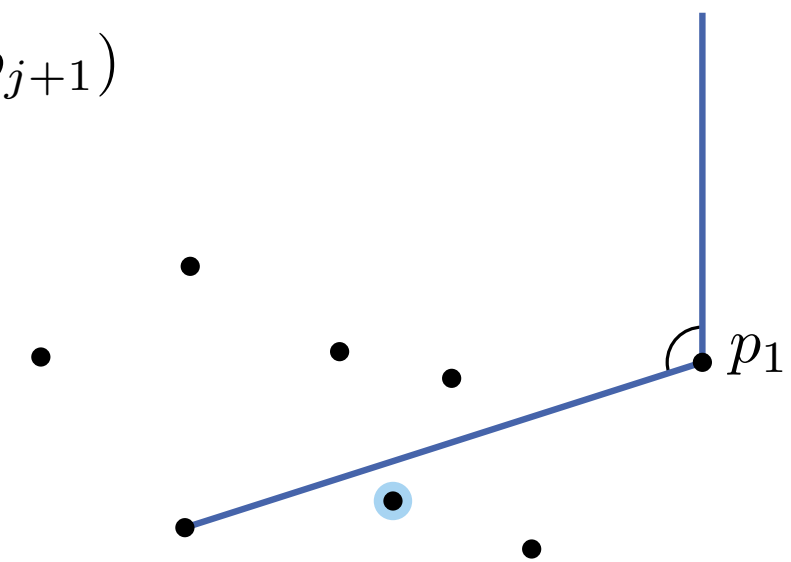

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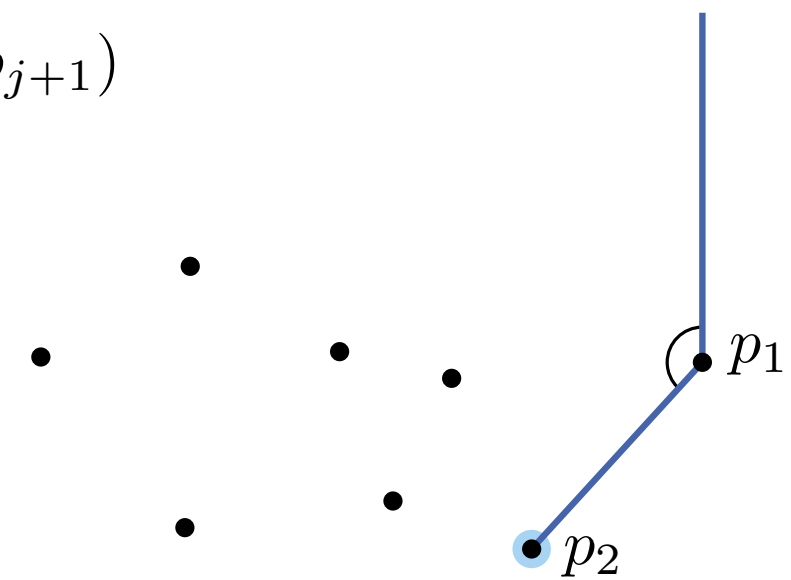

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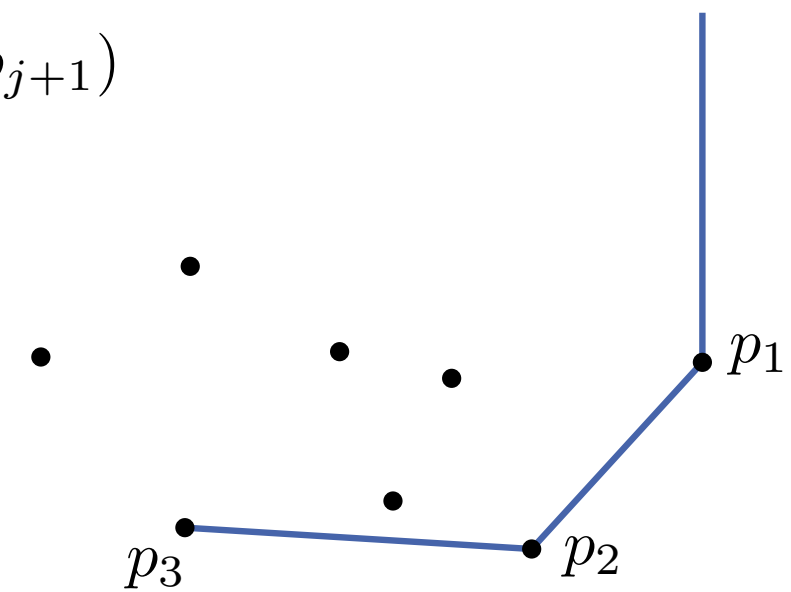

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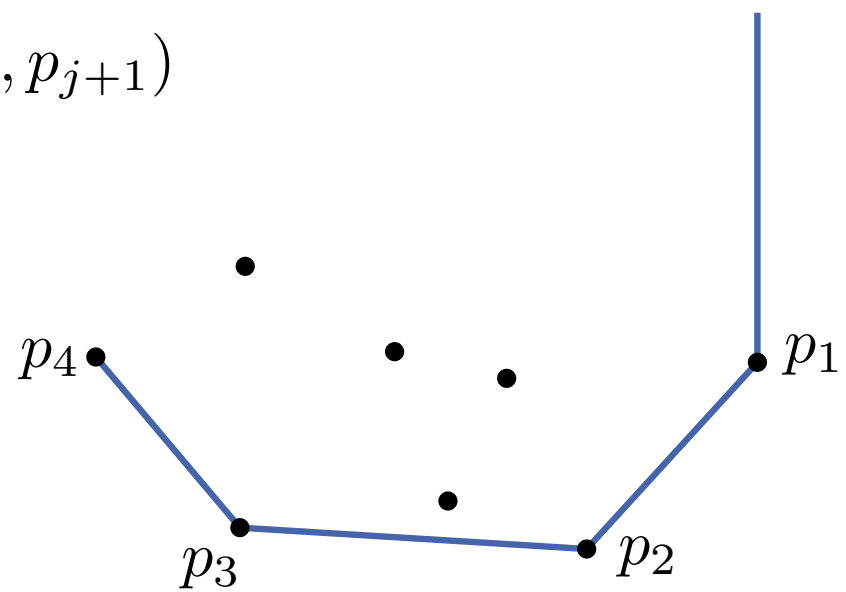

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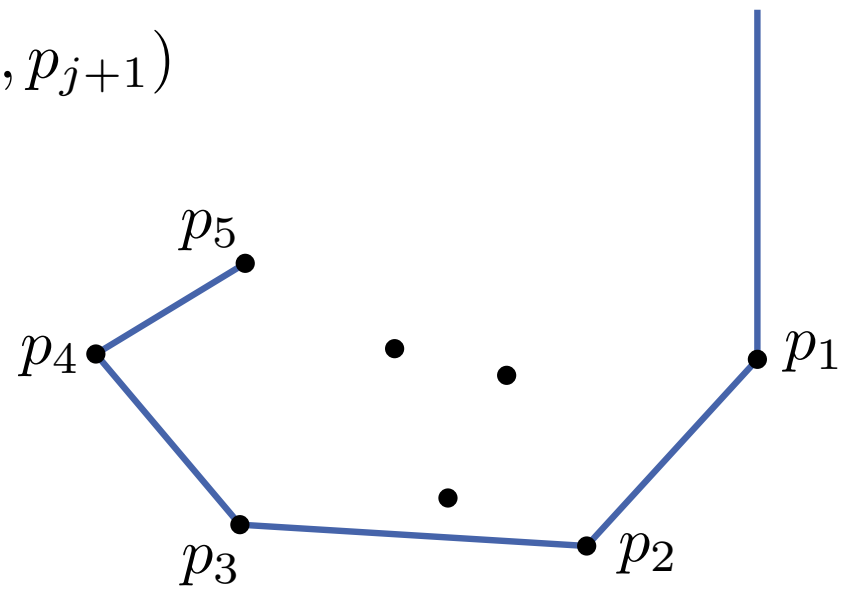

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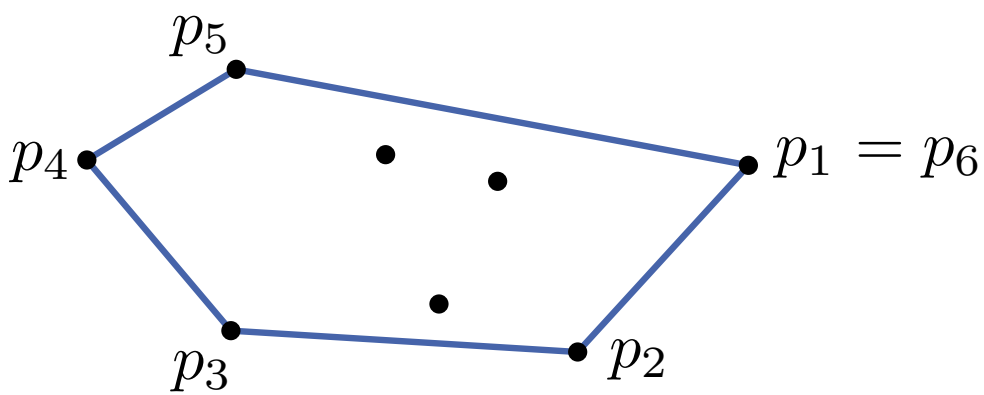

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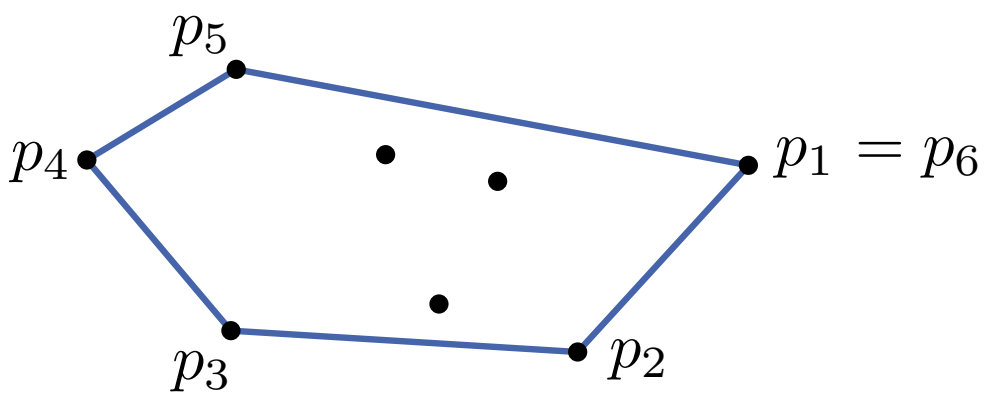

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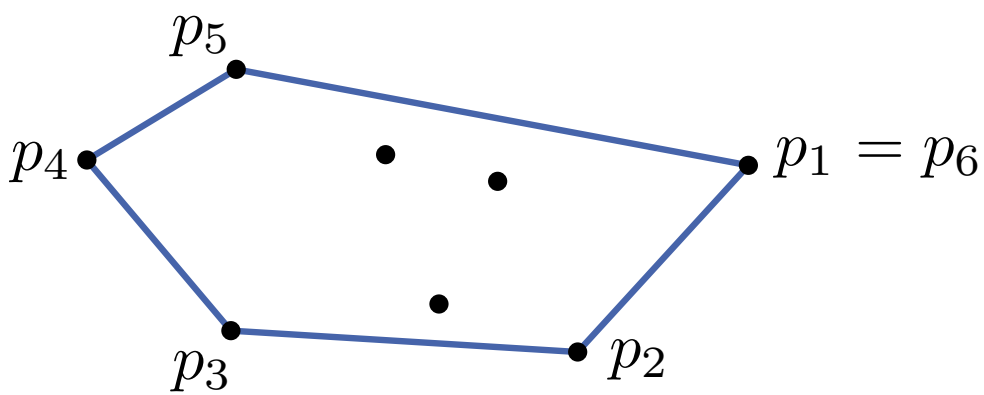

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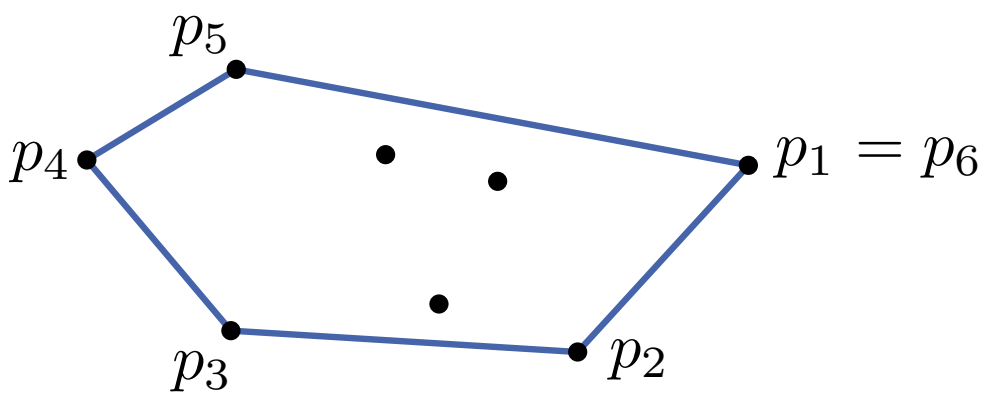

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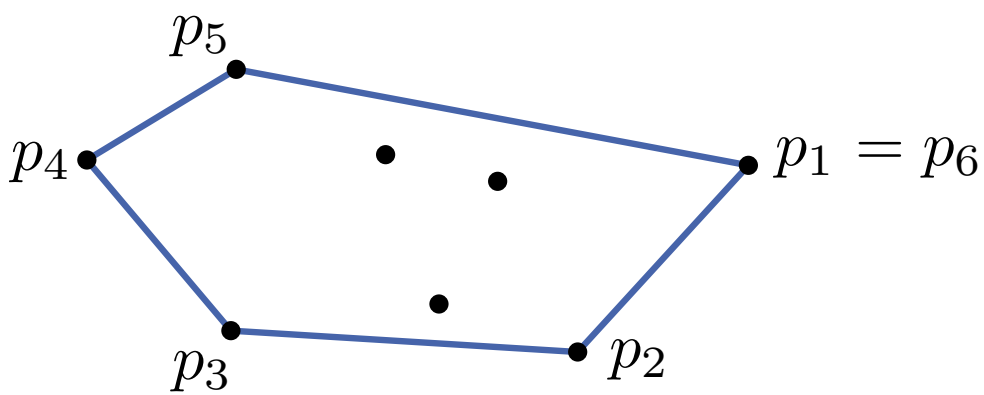

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Theorem. The convex hull $CH(P)$ of n points P in $\mathbb{R}^2$ can be computed in $O(n \cdot h)$ time using *Gift Wrapping* (also called *Jarvis' March*), where $h = |CH(P)|$.

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Proof (sketch): show P is right of $\overrightarrow{p_i p_{i+1}}$ by induction on i

- point p_1 is on CH(P); assume $(p_1, \ldots, p_i)$ are first i points on $CH(P)$
- point p_{i+1} must be right of $\overrightarrow{p_{i-1} p_i}$
- by choice of p_{i+1} all other points are right of $\overrightarrow{p_i p_{i+1}}$

Comparison of the Algorithms

Which algorithm is better?

- Graham's Scan: $O(n \log n)$ time.
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Ques: Can we combine the two approaches into an asymptotically optimal algorithm...?

Chan's Algorithm

Suppose we know h:

ChanHull(P,h)

Divide P into sets P_i with $\leq h$ points

for i from 1 to $\lceil n/h \rceil$ **do**

Compute with GrahamScan $CH(P_i)$

$p_1 = (x_1, y_1) \leftarrow$ rightmost point in P

$p_0 \leftarrow (x_1, \infty)$

for $j = 1$ **to** h **do**

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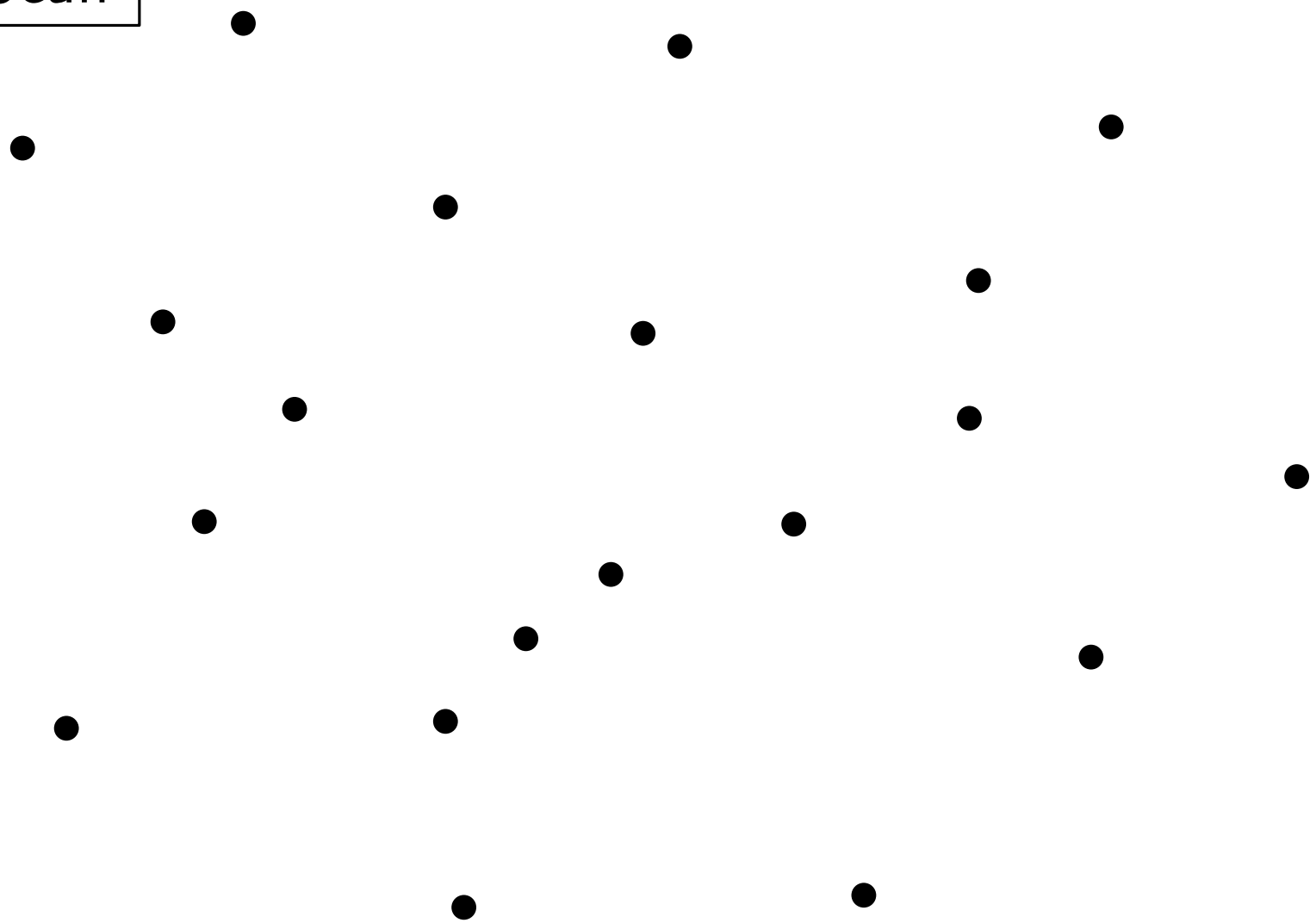
Gift Wrapping

Example

GrahamScan



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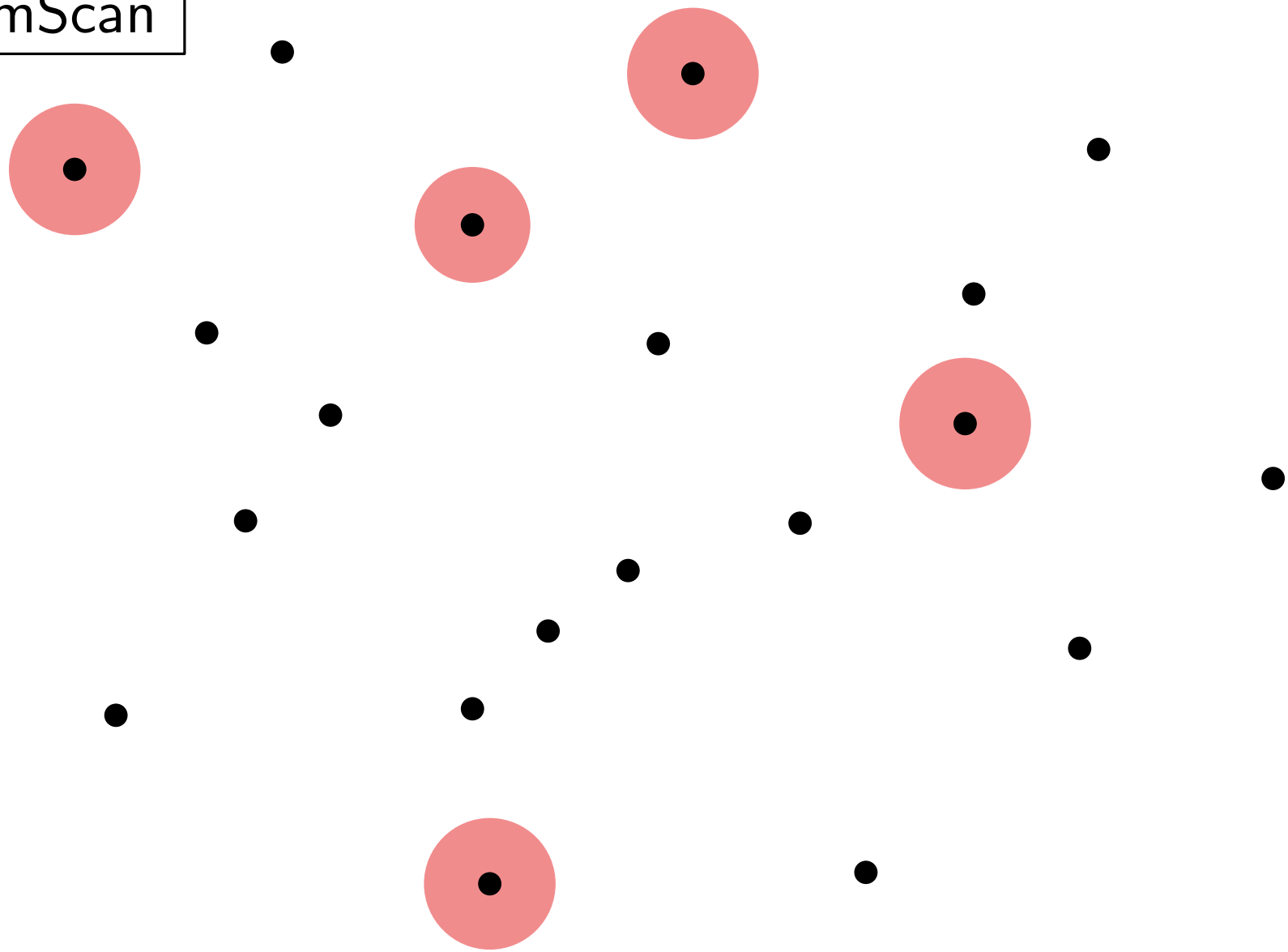

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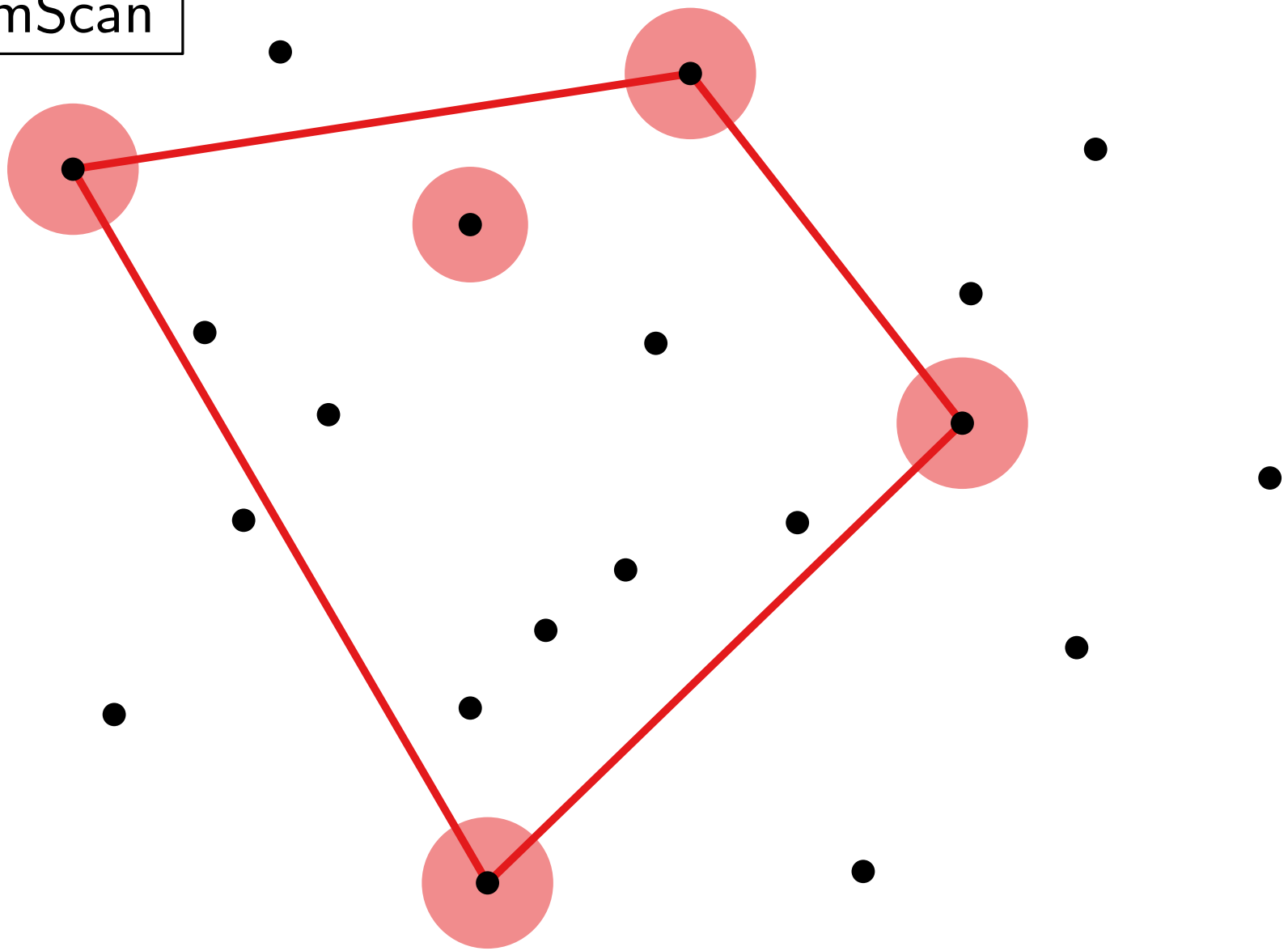
Example

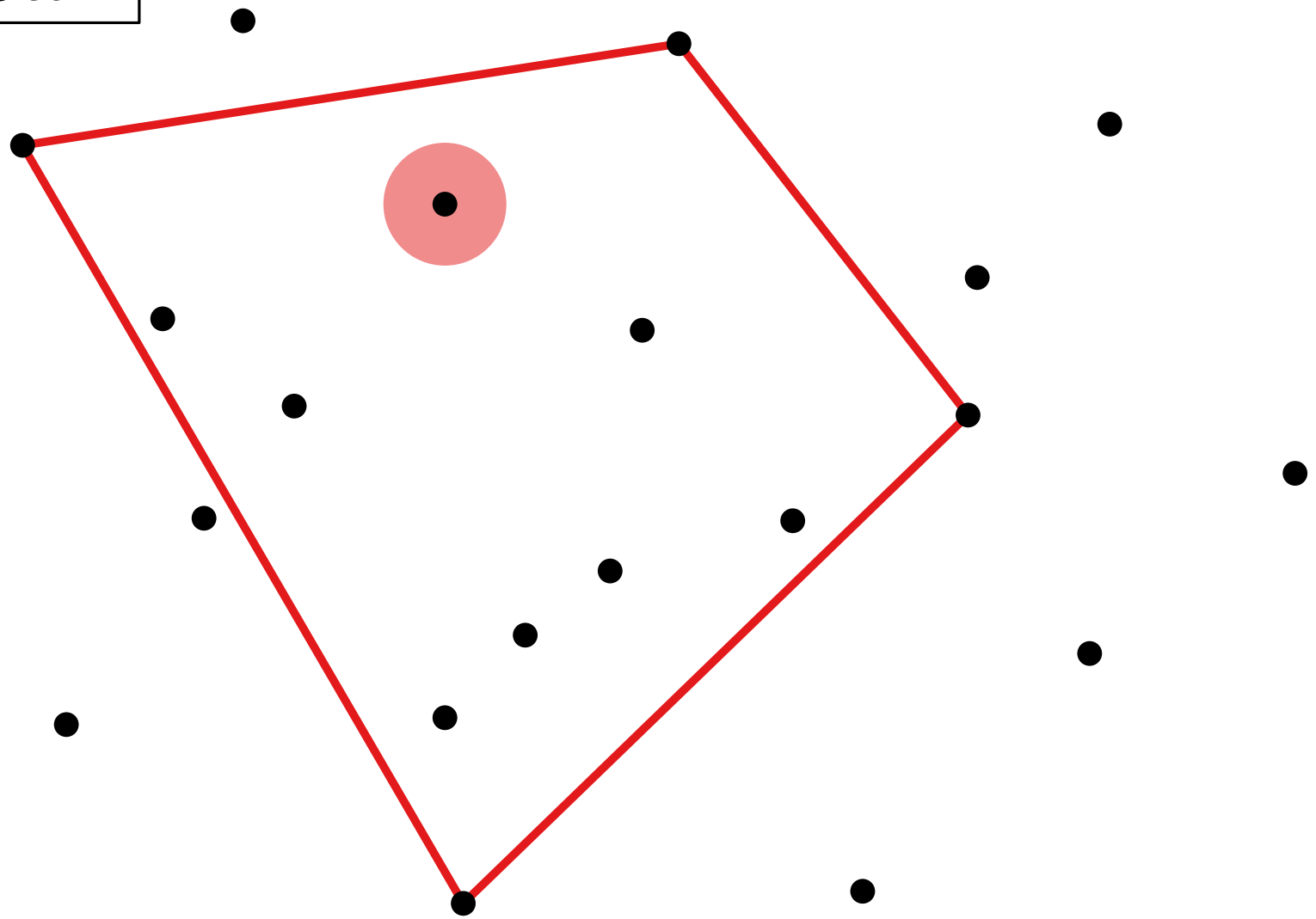
GrahamScan



$n = 20$

Example

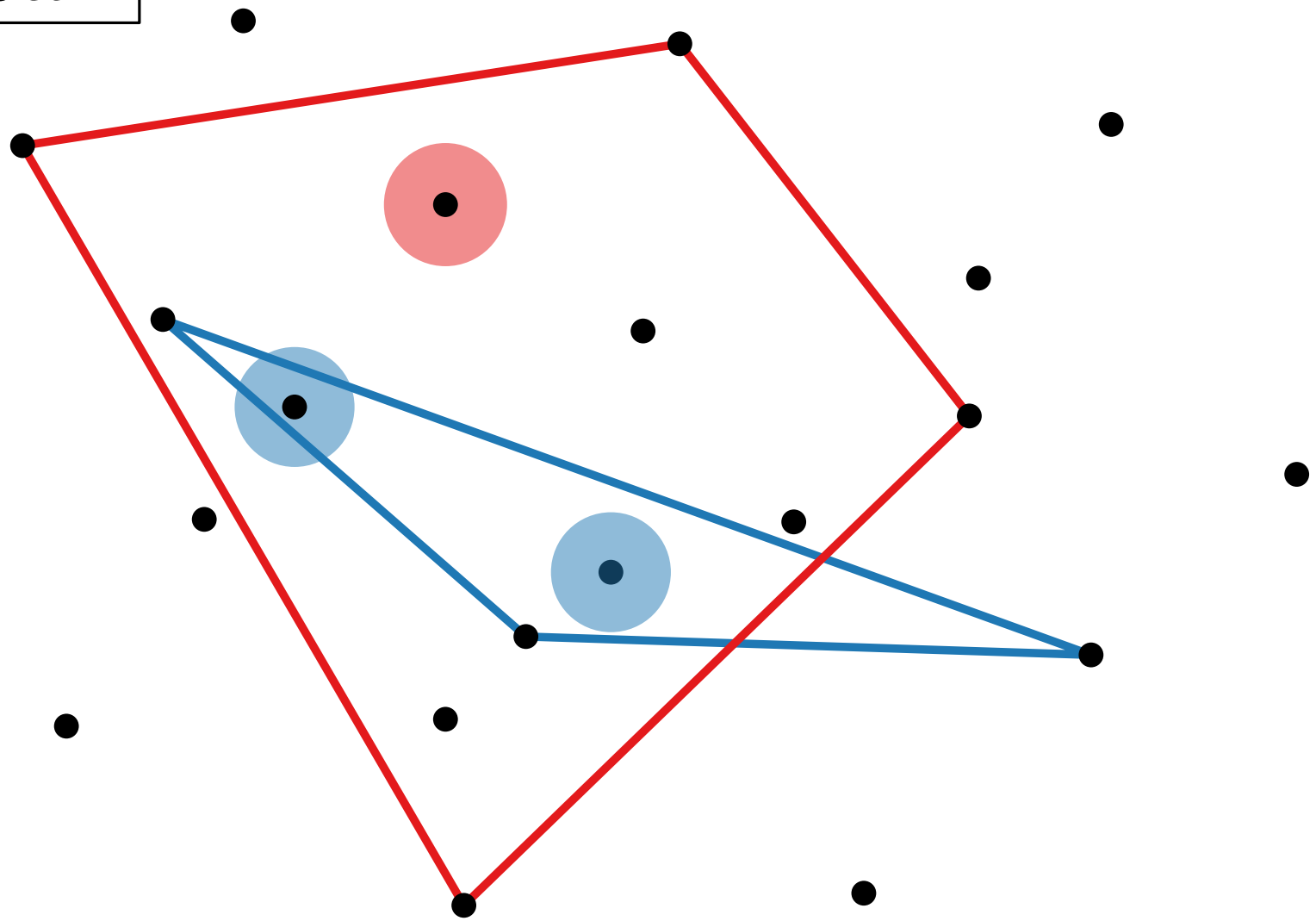
GrahamScan



$n = 20$

Example

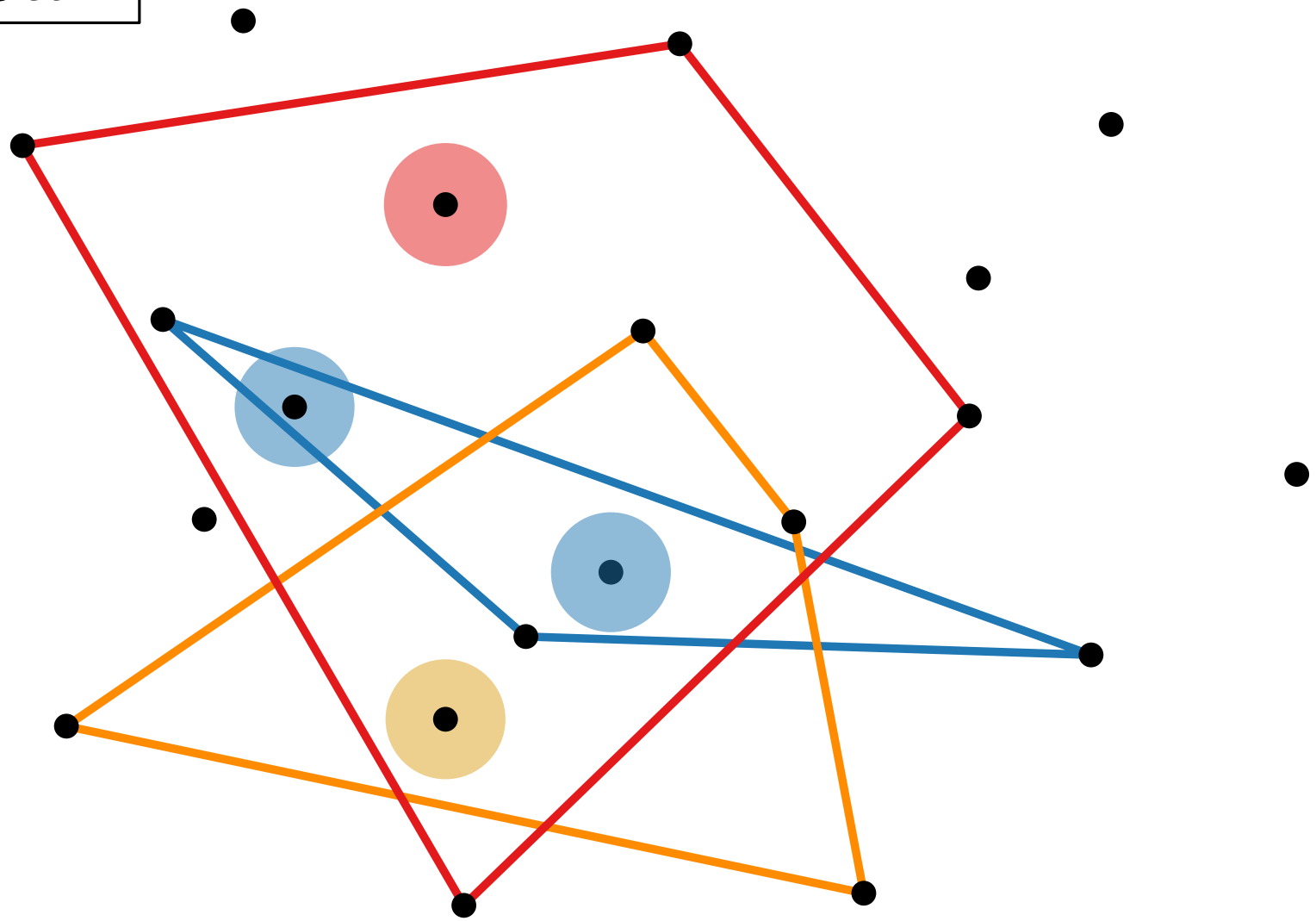
GrahamScan



n = 20

Example

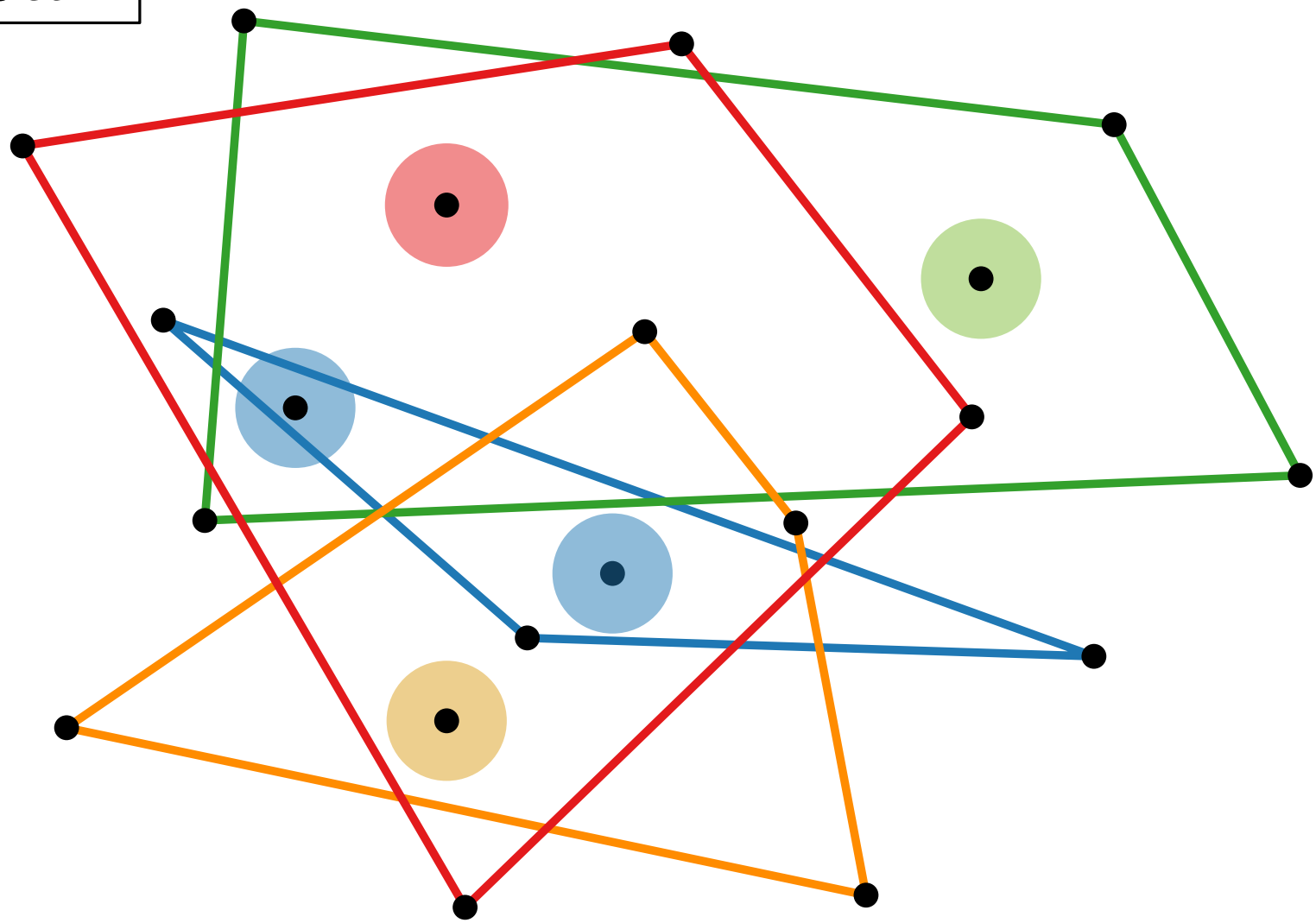
GrahamScan



n = 20

Example

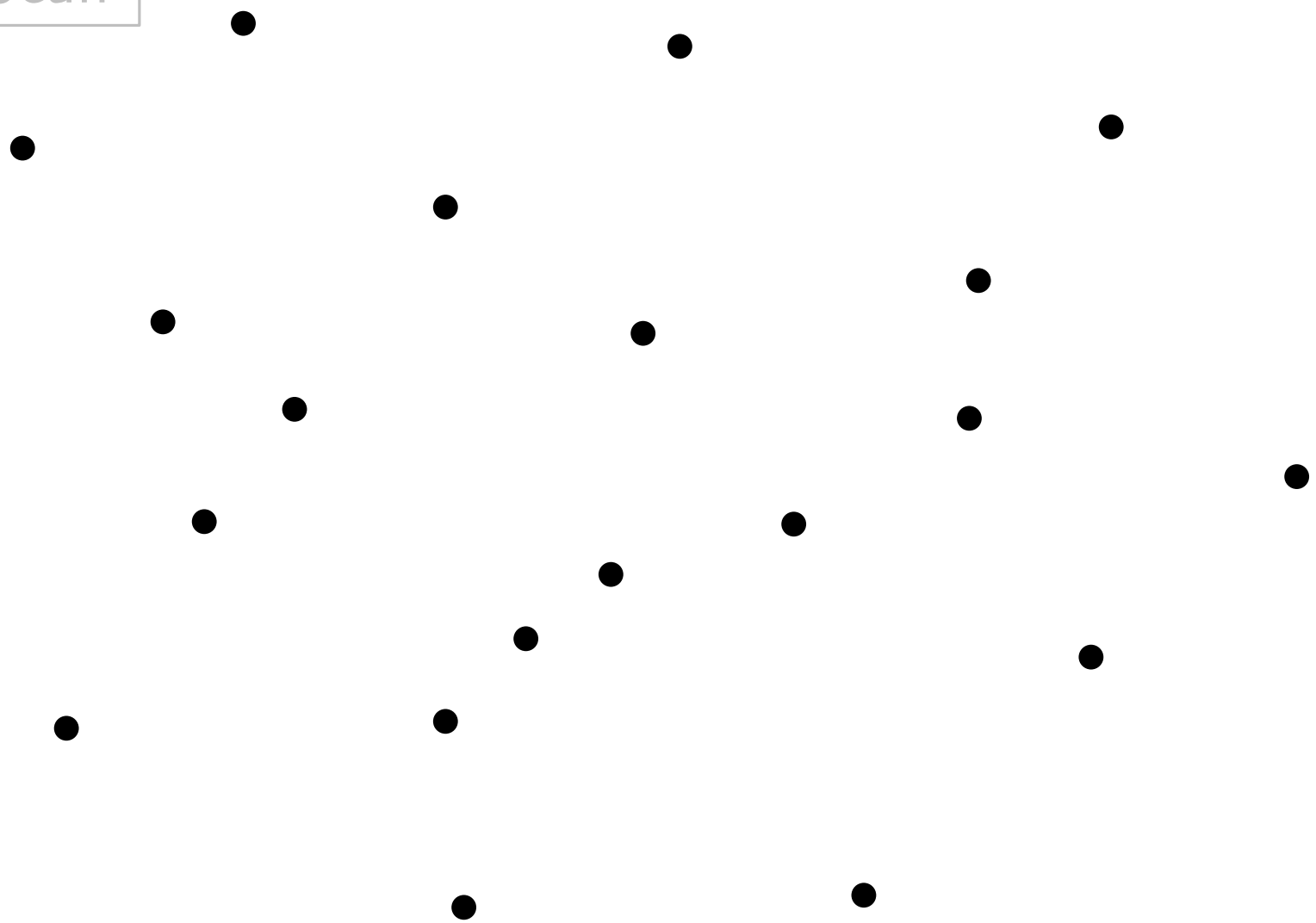
GrahamScan



n = 20

Example

GrahamScan

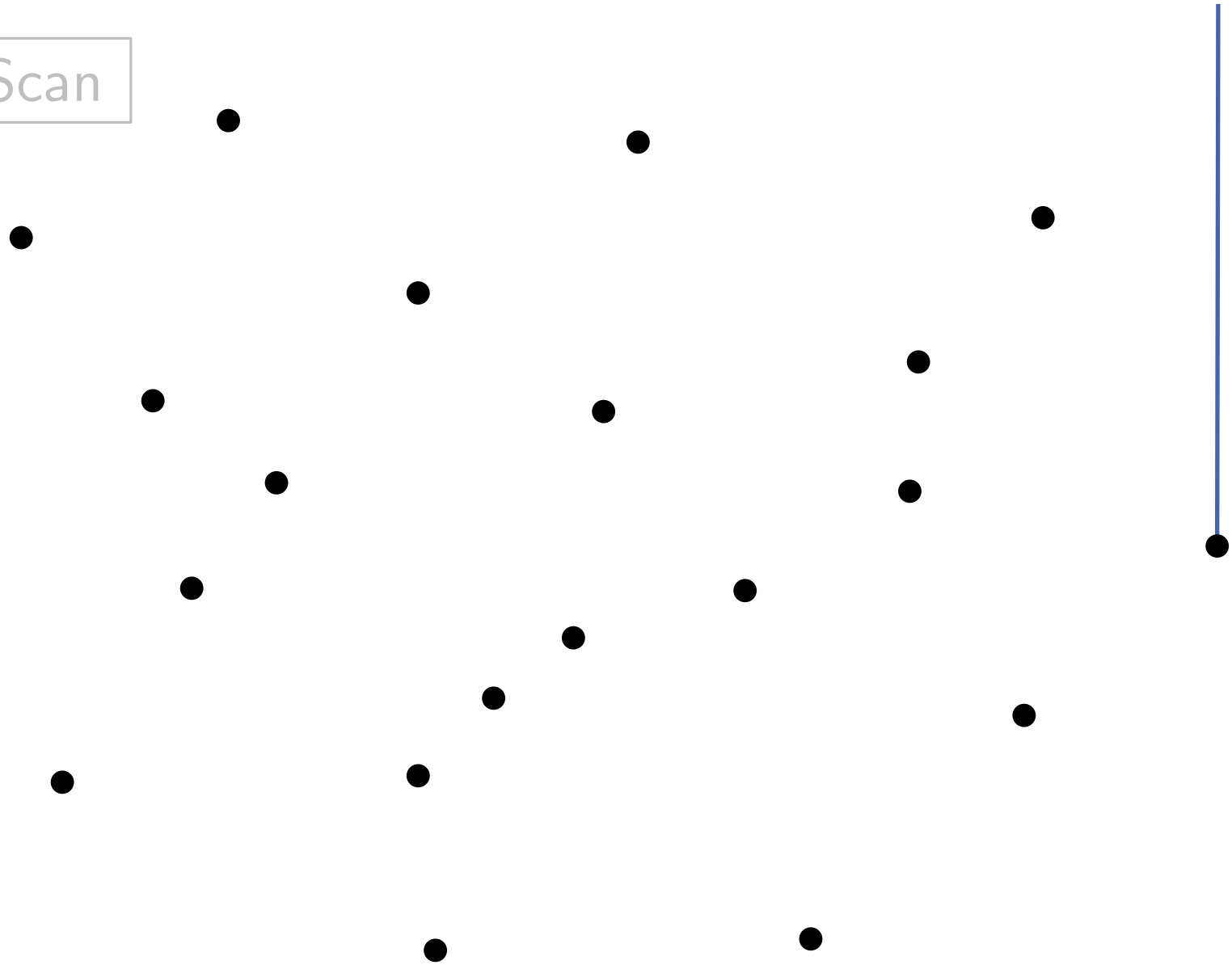


$n = 20$

Gift Wrapping

Example

GrahamScan

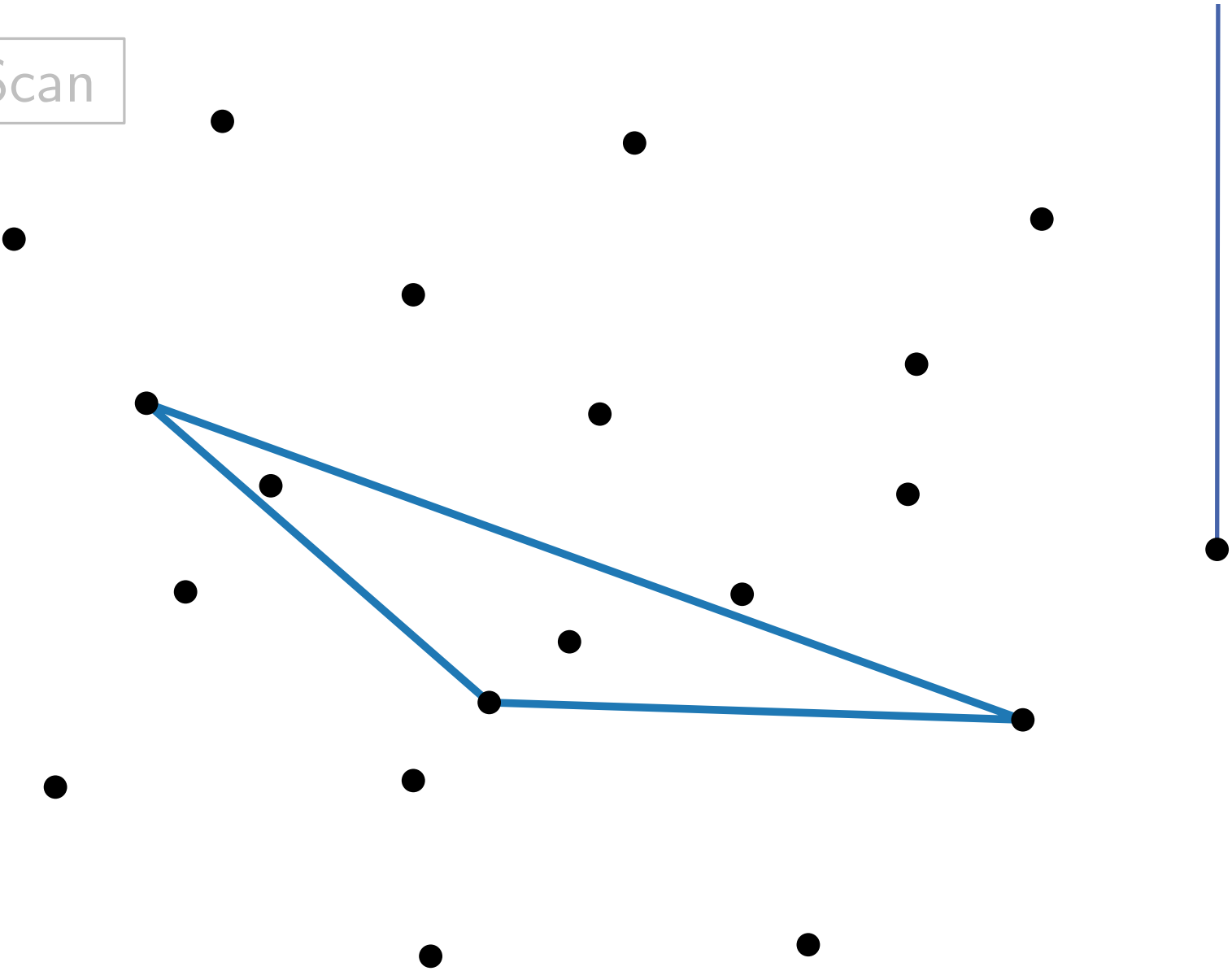


$n = 20$

Gift Wrapping

Example

GrahamScan

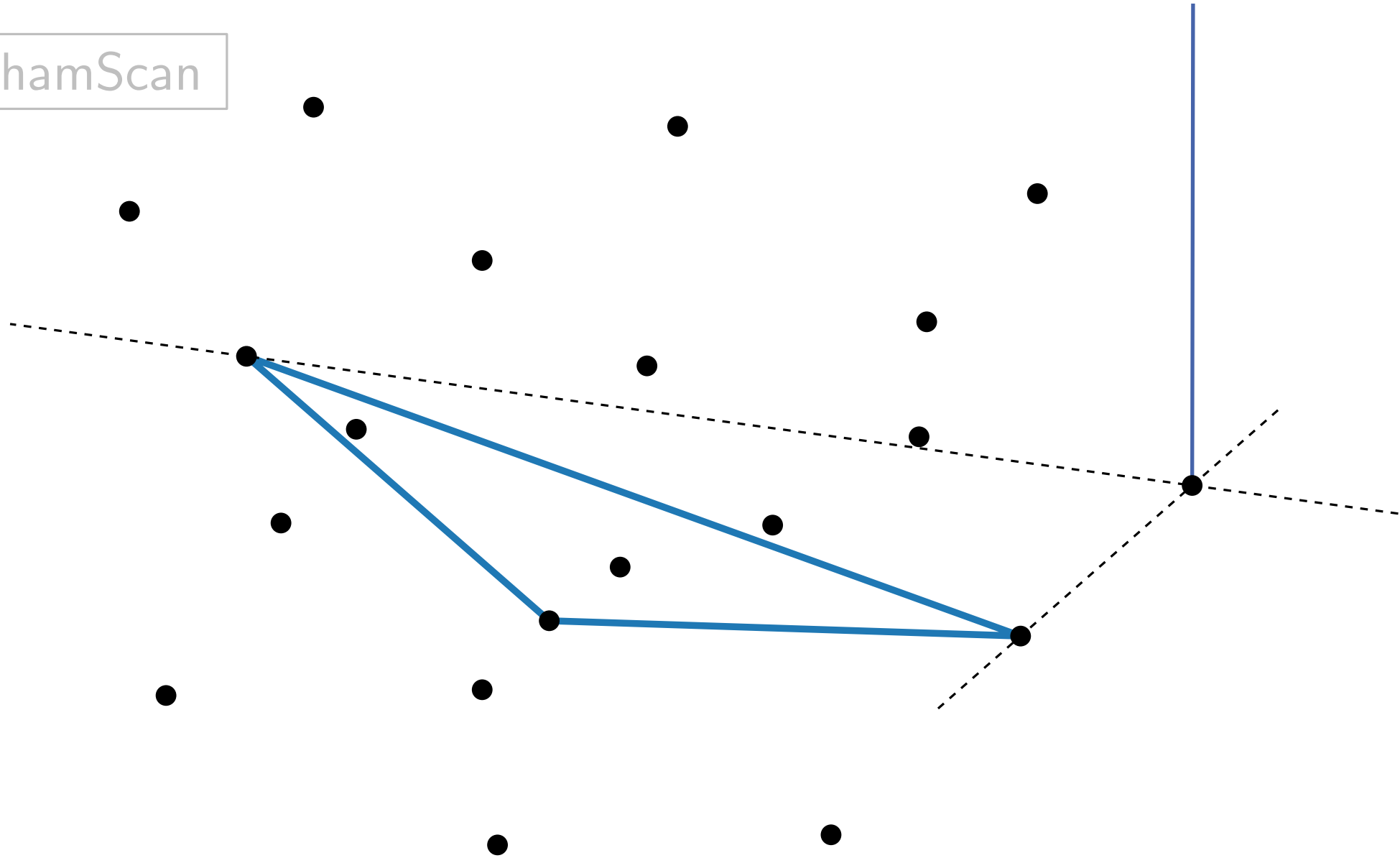


n = 20

Gift Wrapping

Example

GrahamScan

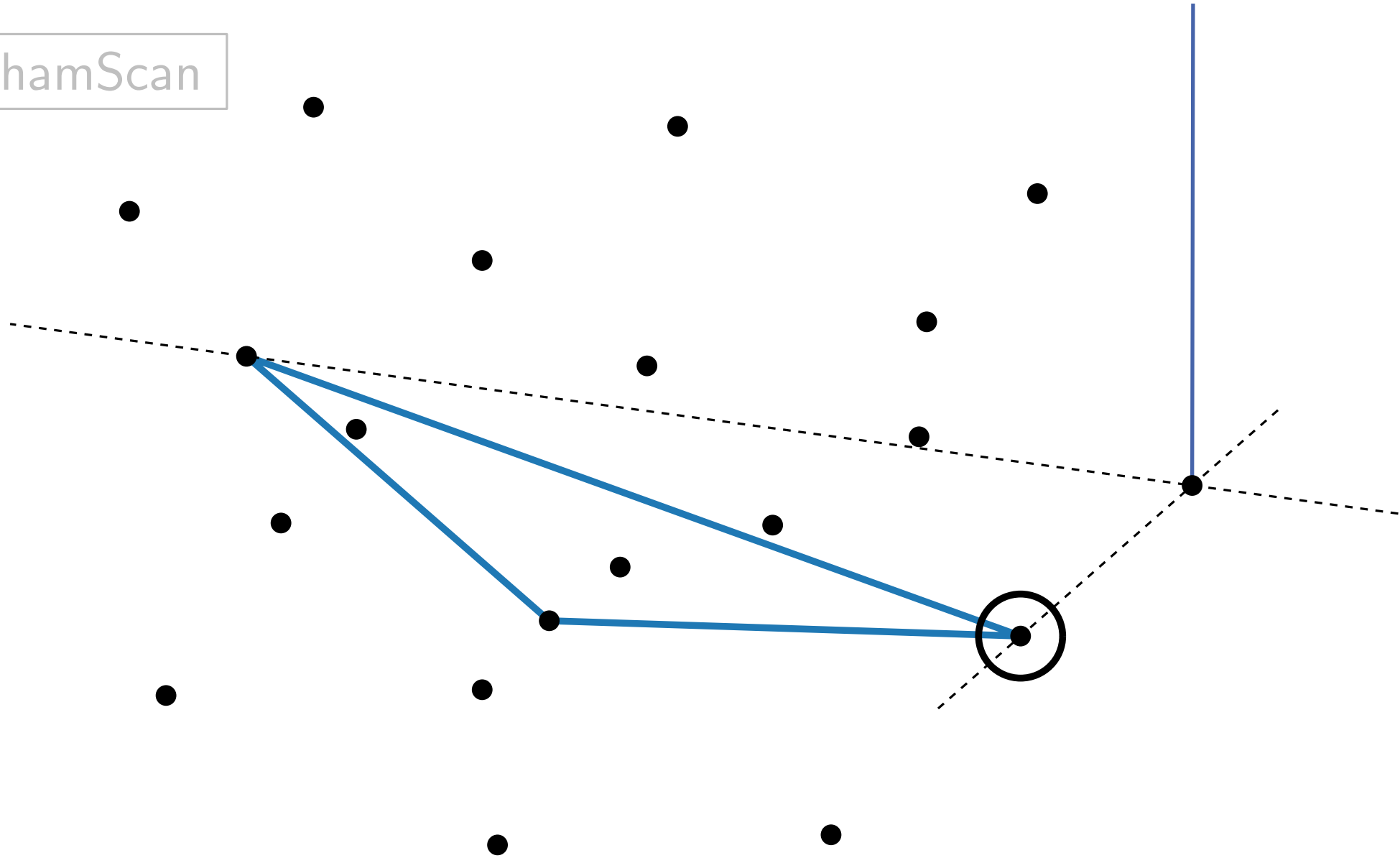


n = 20

Gift Wrapping

Example

GrahamScan

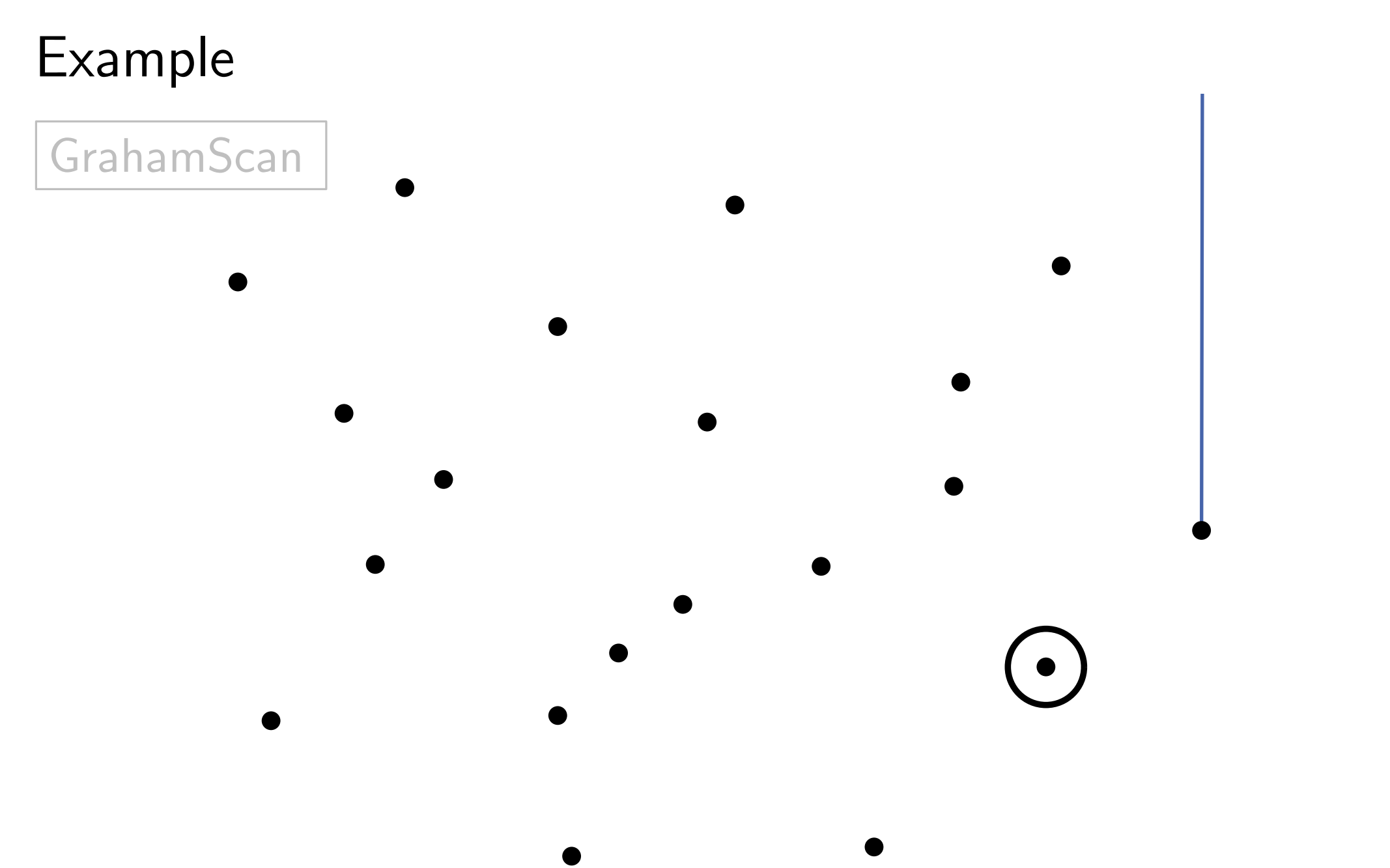


n = 20

Gift Wrapping

Example

GrahamScan

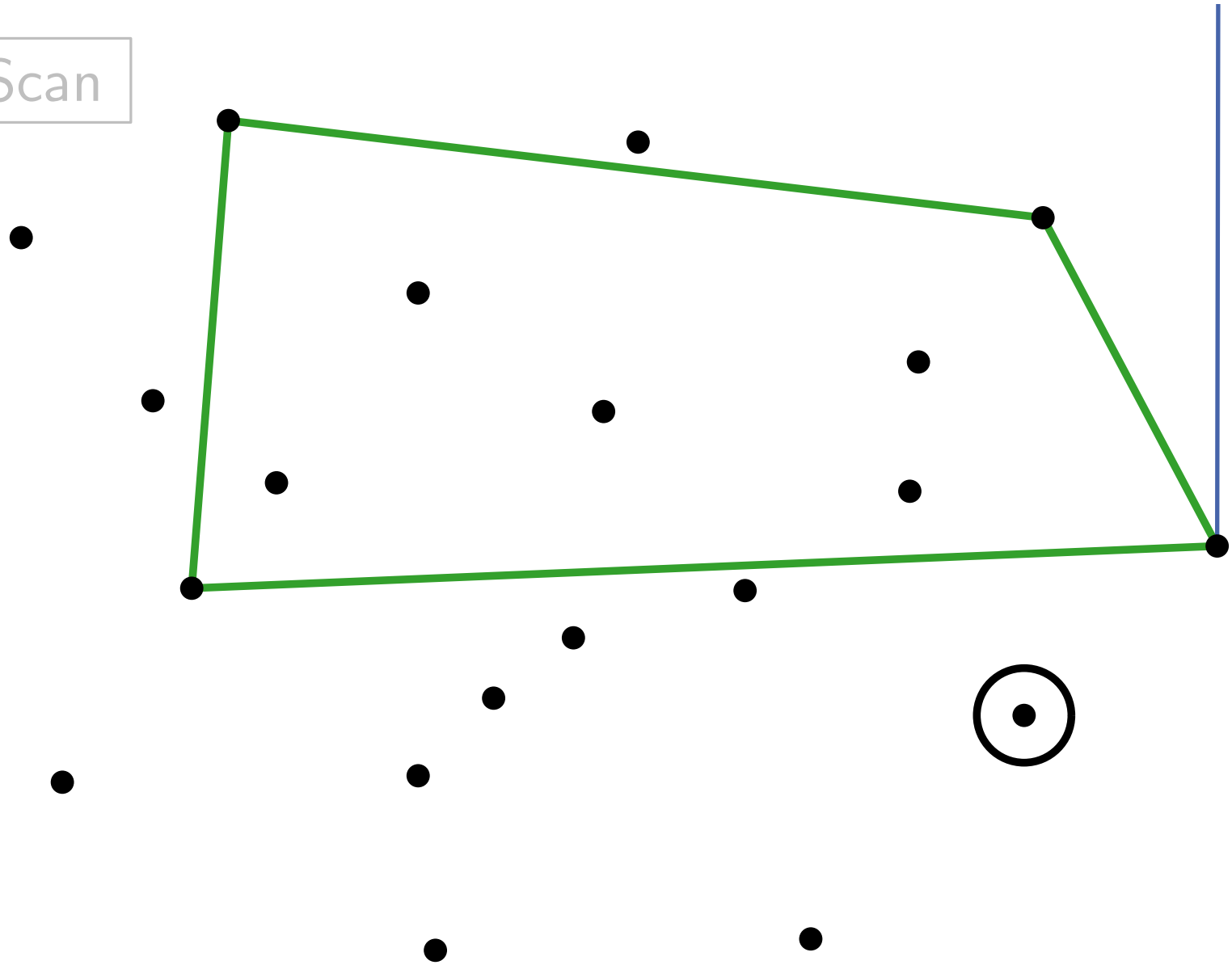


$n = 20$

Gift Wrapping

Example

GrahamScan

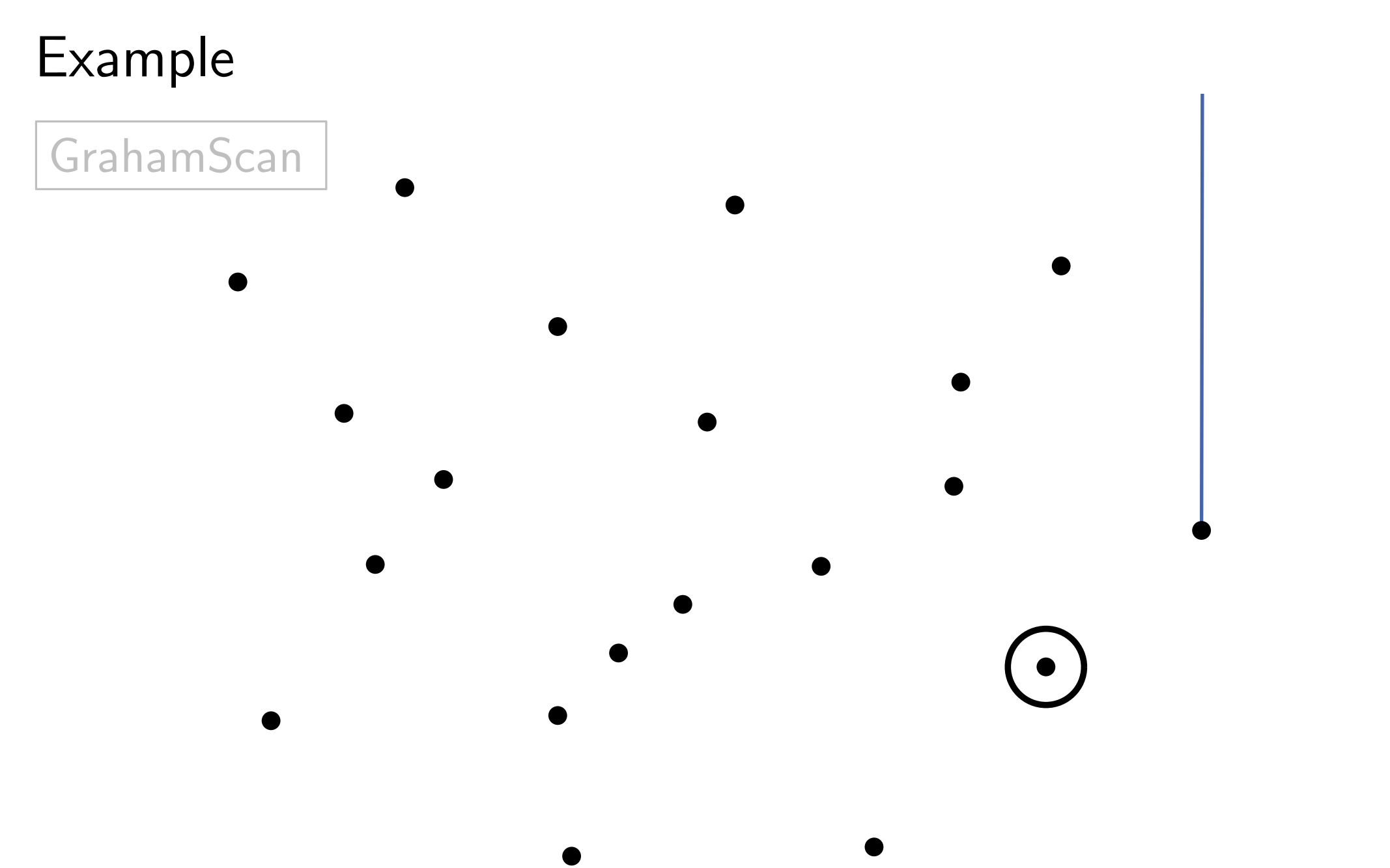


$n = 20$

Gift Wrapping

Example

GrahamScan

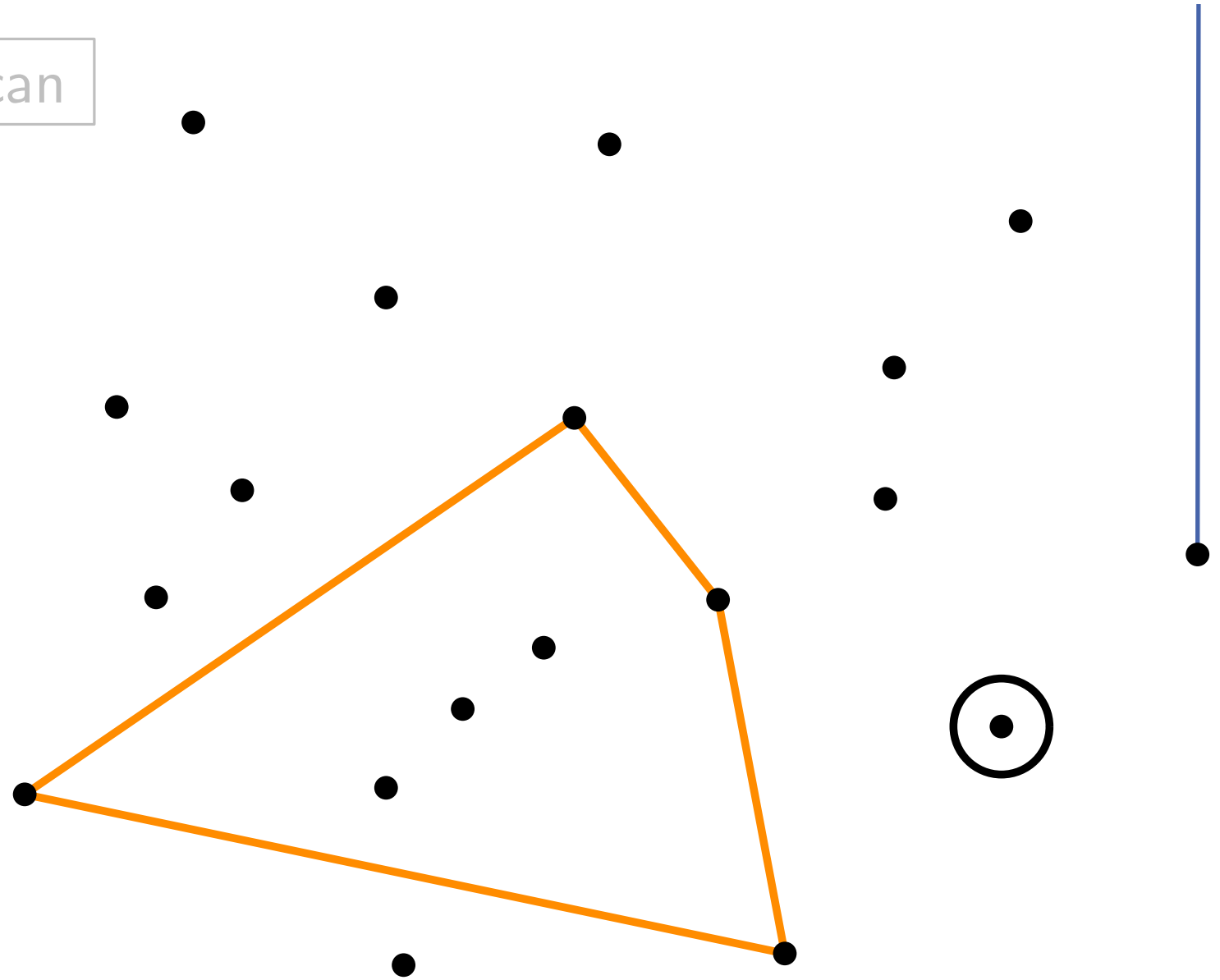


$n = 20$

Gift Wrapping

Example

GrahamScan

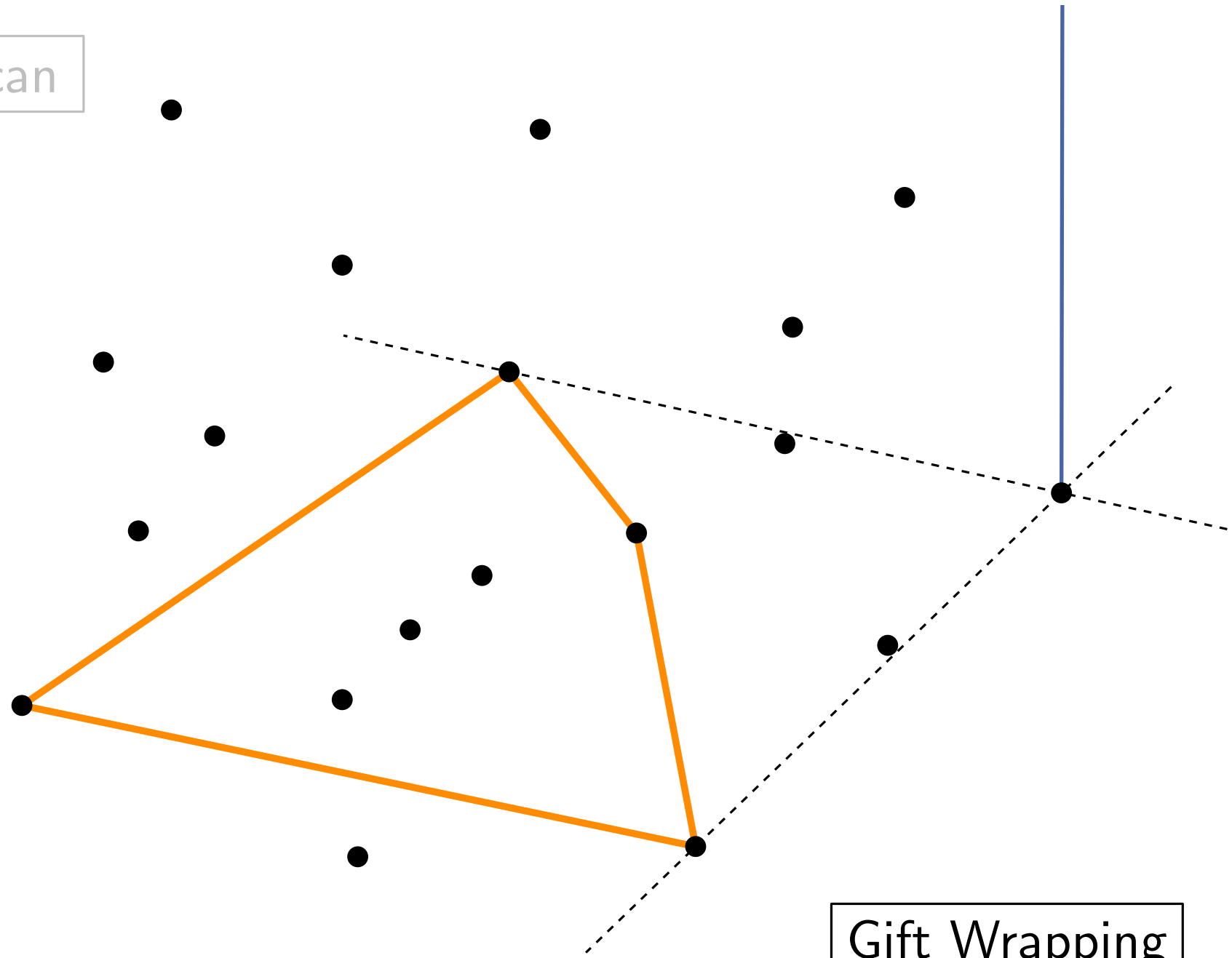


$n = 20$

Gift Wrapping

Example

GrahamScan

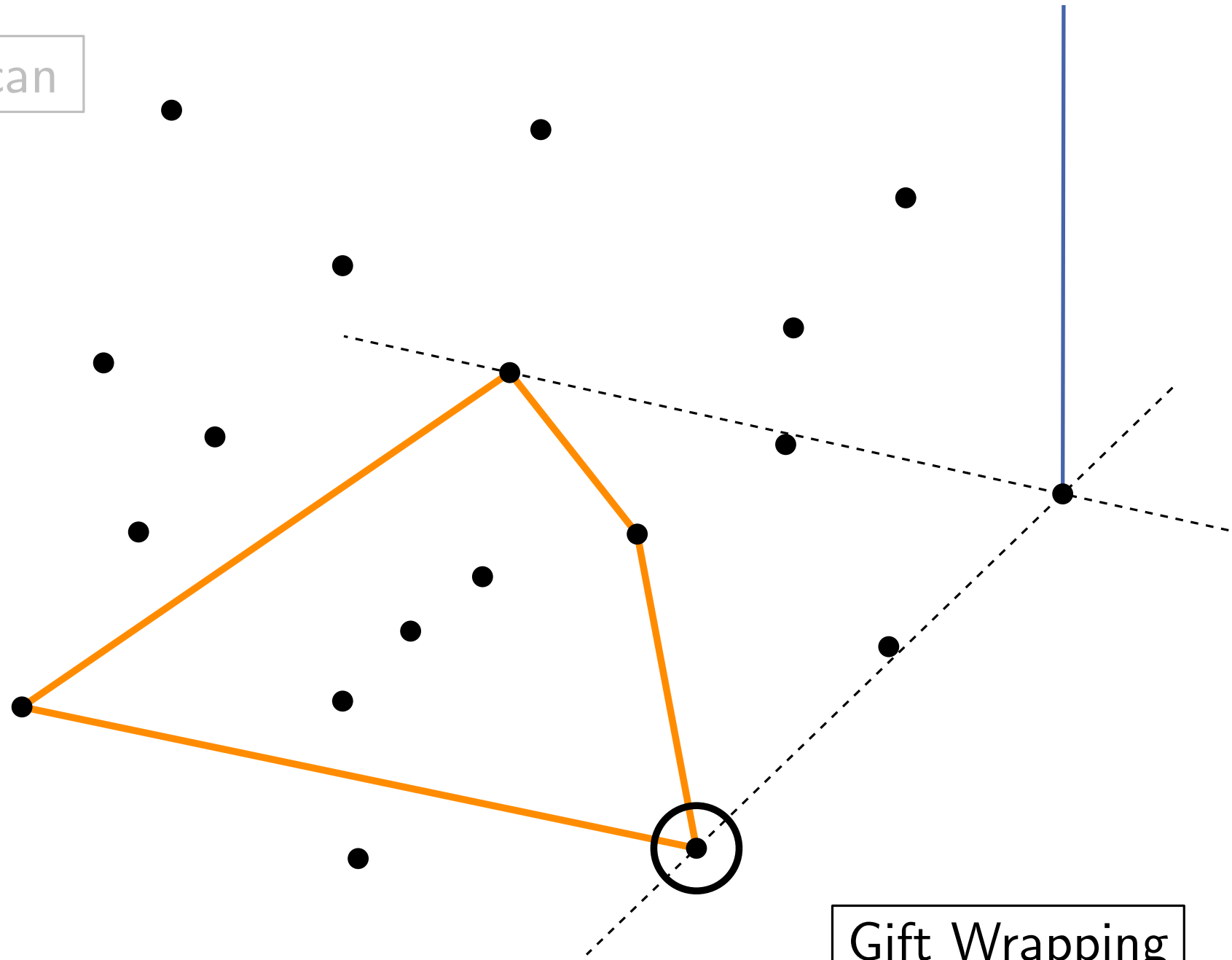


n = 20

Gift Wrapping

Example

GrahamScan

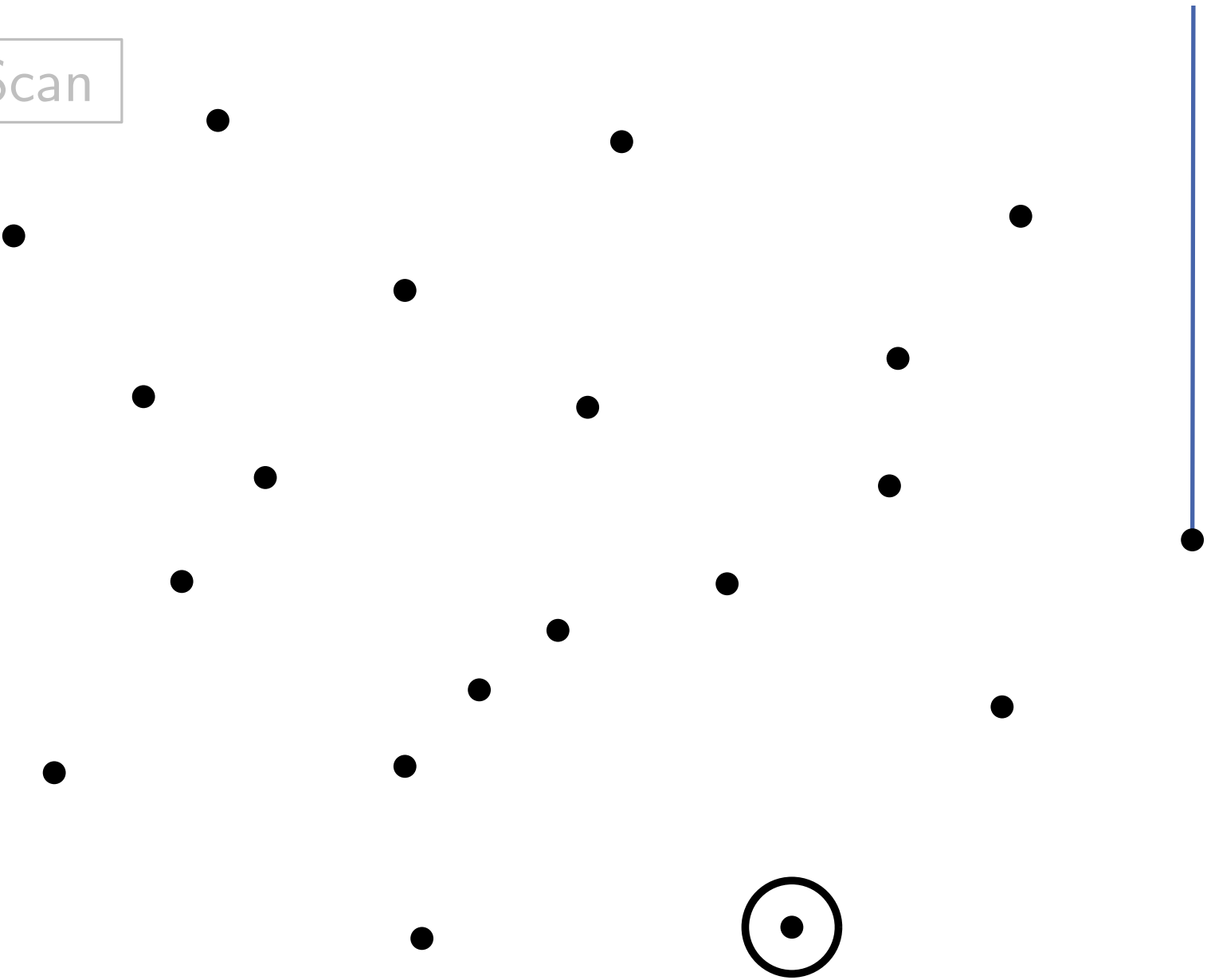


n = 20

Gift Wrapping

Example

GrahamScan

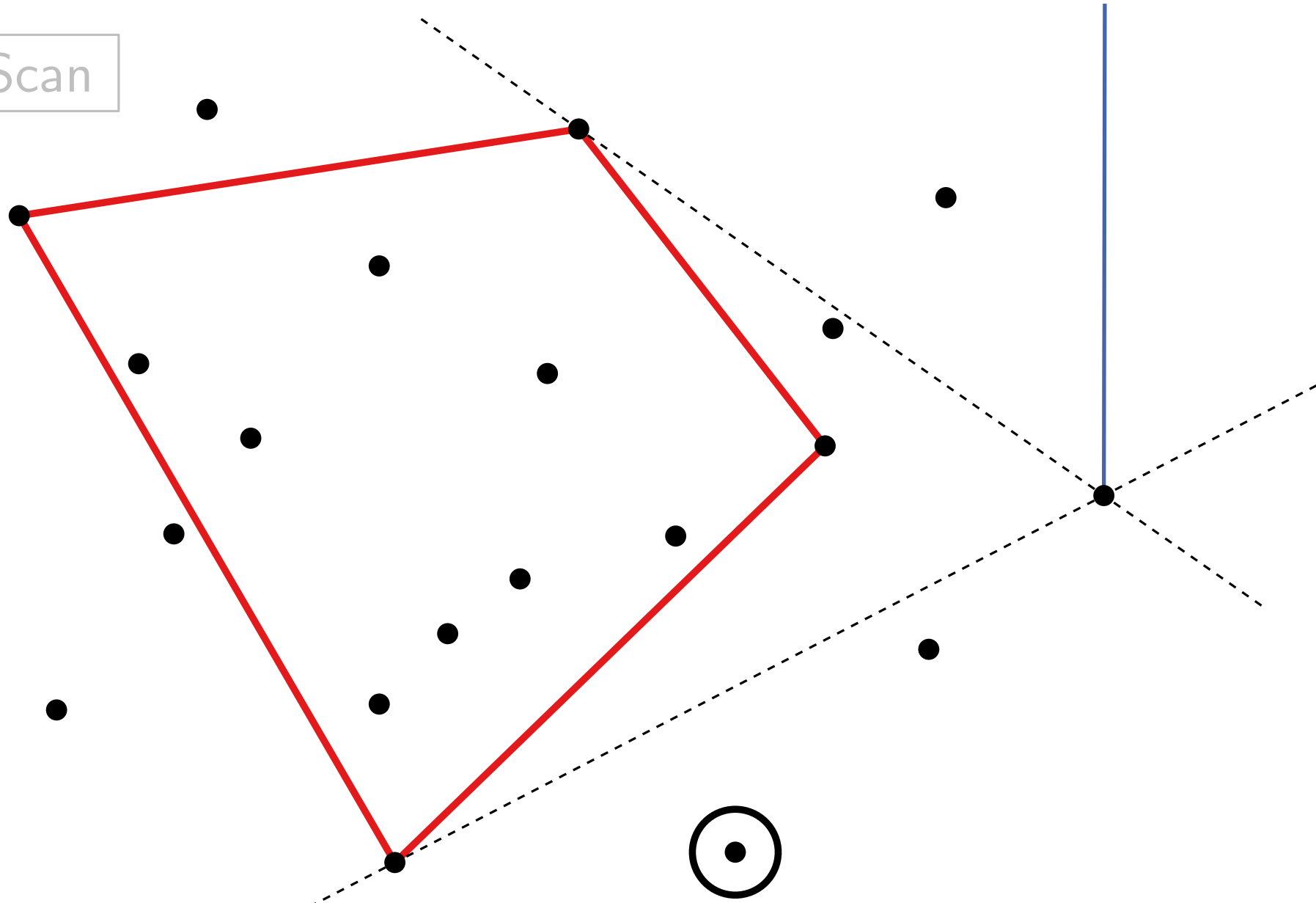


Gift Wrapping

n = 20

Example

GrahamScan

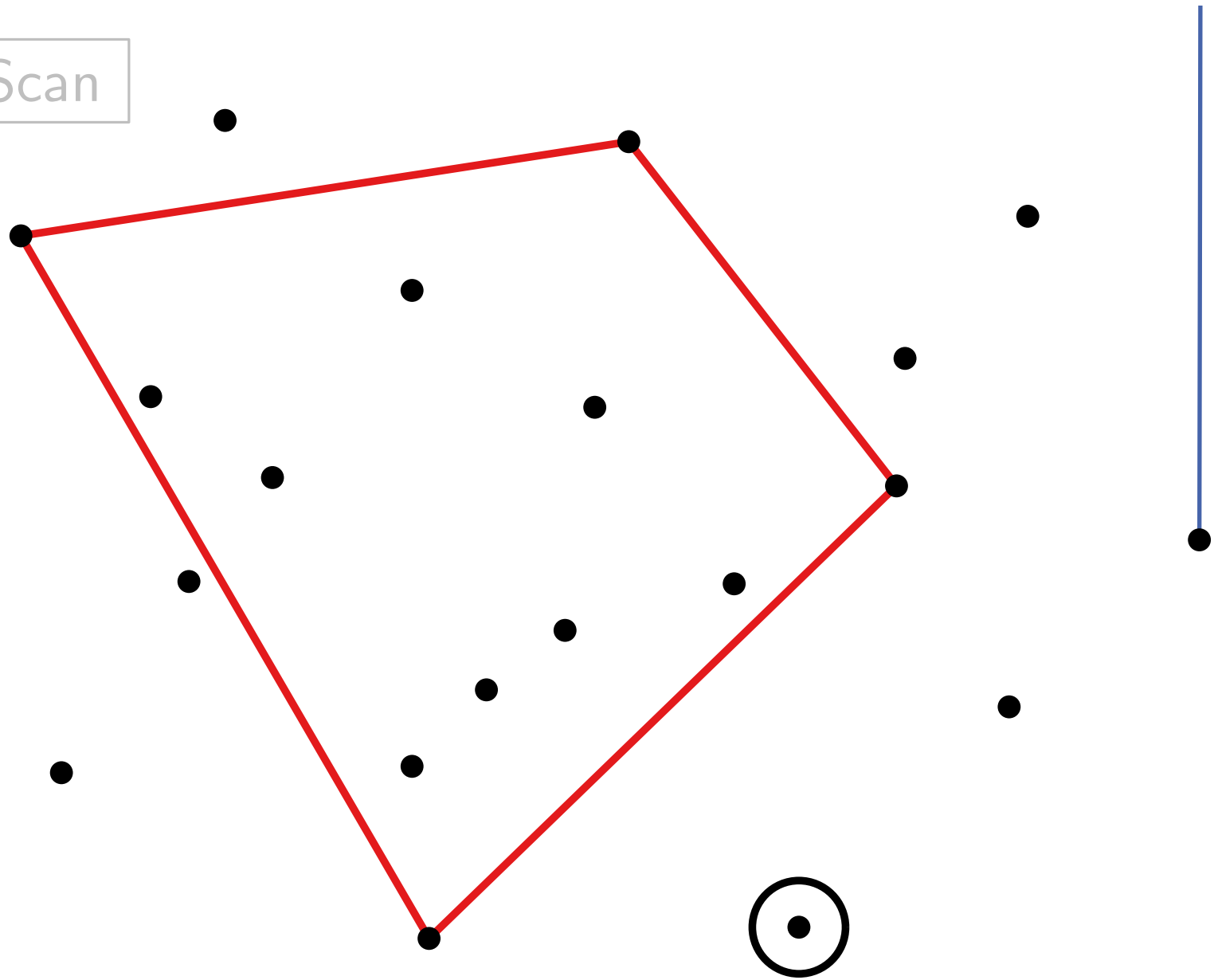


n = 20

Gift Wrapping

Example

GrahamScan

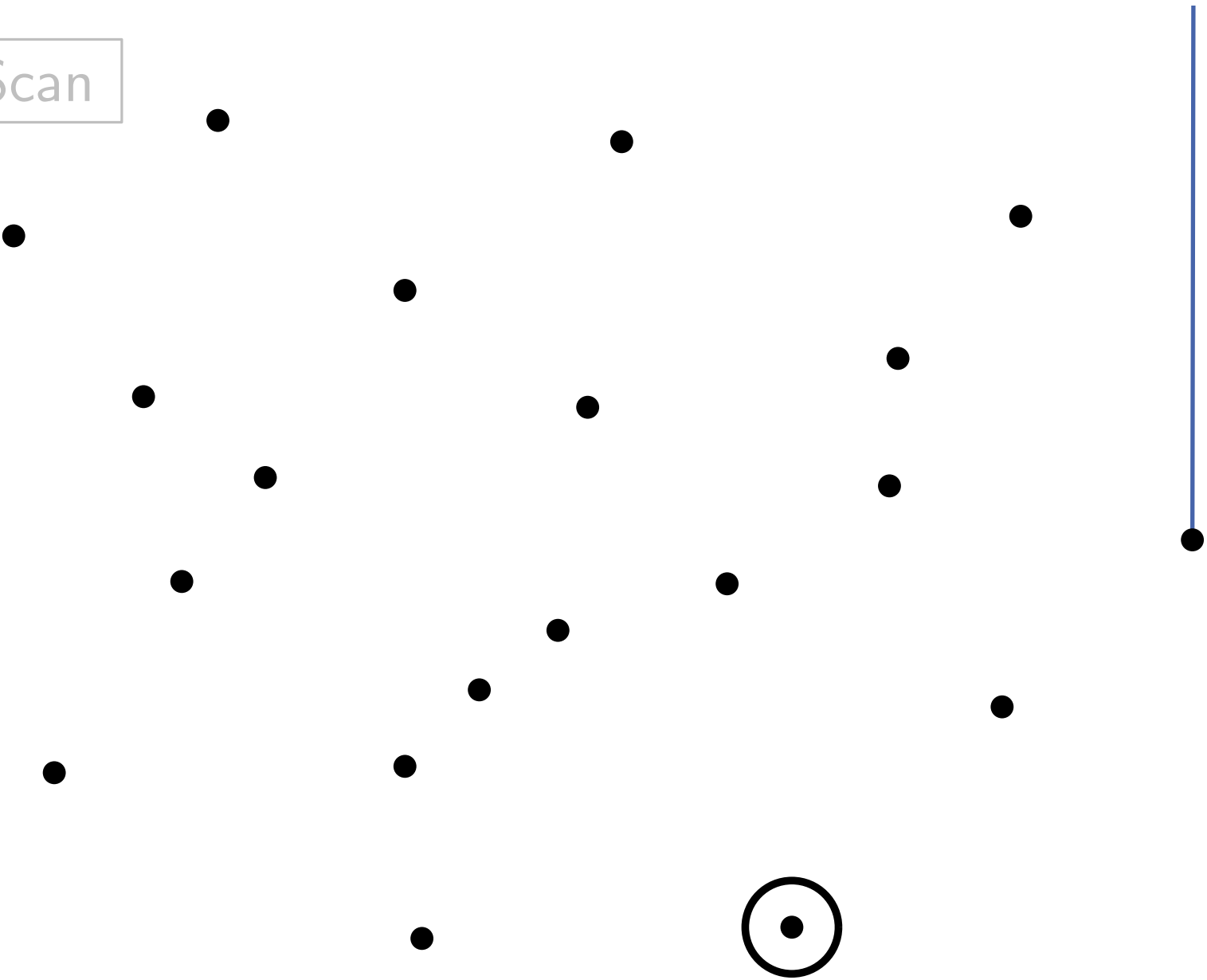


n = 20

Gift Wrapping

Example

GrahamScan

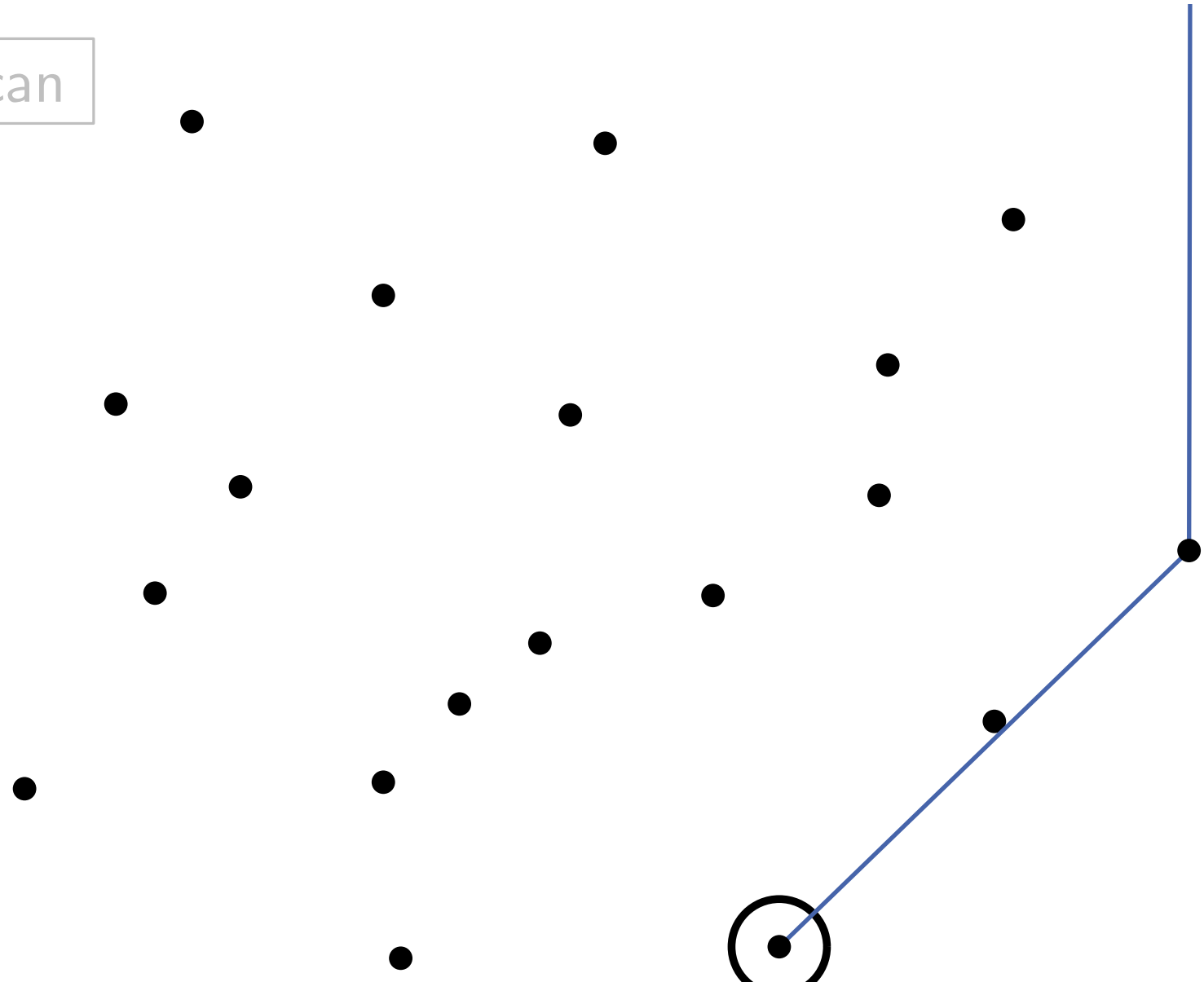


Gift Wrapping

n = 20

Example

GrahamScan



$n = 20$

Gift Wrapping

Chan's Algorithm

Suppose we know h:
ChanHull(P,h)

Divide P into sets P_i with $\leq h$ points

for i from 1 to $\lceil n/h \rceil$ **do**
  Compute with GrahamScan $CH(P_i)$

GrahamScan

$p_1 = (x_1, y_1) \leftarrow$ rightmost point in P
$p_0 \leftarrow (x_1, \infty)$
for $j = 1$ **to** h **do**
  for $i = 1$ **to** $\lceil n/h \rceil$ **do**
    $q_i \leftarrow \arg\max\{\angle p_{j-1} p_j q \mid q \in P_i \setminus \{p_{j-1}, p_j\}\}$
  $p_{j+1} \leftarrow \arg\max\{\angle p_{j-1} p_j q \mid q \in \{q_1, \ldots, q_{\lceil n/h \rceil}\}\}$
return $(p_1, \ldots, p_h)$

Gift Wrapping

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for i from 1 to $\lceil n/h \rceil$ **do**
  Compute with GrahamScan $CH(P_i)$ $\mathcal{O}(h \log h)$

$p_1 = (x_1, y_1) \leftarrow$ rightmost point in P
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$\mathcal{O}(\log h) \rightarrow$ exercise-1

  $p_{j+1} \leftarrow \arg\max\{\angle p_{j-1} p_j q \mid q \in \{q_1, \ldots, q_{\lceil n/h \rceil}\}\}$

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$\mathcal{O}(n \log h)$

$p_1 = (x_1, y_1) \leftarrow$ rightmost point in P

$p_0 \leftarrow (x_1, \infty)$

$\mathcal{O}(h) \cdot \mathcal{O}(n/h) = \mathcal{O}(n)$

for $j = 1$ **to** h **do**

  for $i = 1$ **to** $\lceil n/h \rceil$ **do** $\mathcal{O}(\log h) \rightarrow$ exercise-1

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Divide P into sets P_i with $\leq h$ points

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$\mathcal{O}(n \log h)$

return $(p_1, \ldots, p_h)$

Chan's Algorithm

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ChanHull(P,h)

Divide P into sets P_i with $\leq h$ points

for i from 1 to $\lceil n/h \rceil$ **do** $\mathcal{O}(n/h)$

  Compute with GrahamScan $CH(P_i)$ $\mathcal{O}(h \log h)$

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  for $i = 1$ **to** $\lceil n/h \rceil$ **do** $\mathcal{O}(\log h) \rightarrow$ exercise-1

    $q_i \leftarrow \arg\max\{\angle p_{j-1} p_j q \mid q \in P_i \setminus \{p_{j-1}, p_j\}\}$

  $p_{j+1} \leftarrow \arg\max\{\angle p_{j-1} p_j q \mid q \in \{q_1, \ldots, q_{\lceil n/h \rceil}\}\}$

$\mathcal{O}(n \log h)$

return $(p_1, \ldots, p_h)$

Total: $\mathcal{O}(n \log h)$

Chan's Algorithm

But in general h is unknown!

ChanHull(P,h)

Divide P into sets P_i with $\leq h$ points

for i from 1 to $\lceil n/h \rceil$ **do** $\mathcal{O}(n/h)$

  Compute with GrahamScan $CH(P_i)$ $\mathcal{O}(h \log h)$

$\mathcal{O}(n \log h)$

$p_1 = (x_1, y_1) \leftarrow$ rightmost point in P

$p_0 \leftarrow (x_1, \infty)$

$\mathcal{O}(h) \cdot \mathcal{O}(n/h) = \mathcal{O}(n)$

for $j = 1$ **to** h **do**

  for $i = 1$ **to** $\lceil n/h \rceil$ **do**

    $q_i \leftarrow \arg\max\{\angle p_{j-1} p_j q \mid q \in P_i \setminus \{p_{j-1}, p_j\}\}$

$\mathcal{O}(\log h) \rightarrow$ exercise-1

  $p_{j+1} \leftarrow \arg\max\{\angle p_{j-1} p_j q \mid q \in \{q_1, \ldots, q_{\lceil n/h \rceil}\}\}$

$\mathcal{O}(n \log h)$

return $(p_1, \ldots, p_h)$

Chan's Algorithm

But in general h is unknown!

ChanHull(P, ~~h~~ m)

Divide P into sets P_i with $\leq h$ points

for i from 1 to $\lceil n/h \rceil$ **do** $\mathcal{O}(n/h)$

  Compute with GrahamScan $CH(P_i)$ $\mathcal{O}(h \log h)$

$\mathcal{O}(n \log h)$

$p_1 = (x_1, y_1) \leftarrow$ rightmost point in P

$p_0 \leftarrow (x_1, \infty)$

$\mathcal{O}(h) \cdot \mathcal{O}(n/h) = \mathcal{O}(n)$

for $j = 1$ **to** h **do**

  for $i = 1$ **to** $\lceil n/h \rceil$ **do** $\mathcal{O}(\log h) \rightarrow$ exercise-1

    $q_i \leftarrow \arg\max\{\angle p_{j-1} p_j q \mid q \in P_i \setminus \{p_{j-1}, p_j\}\}$

  $p_{j+1} \leftarrow \arg\max\{\angle p_{j-1} p_j q \mid q \in \{q_1, \ldots, q_{\lceil n/h \rceil}\}\}$

$\mathcal{O}(n \log h)$

return $(p_1, \ldots, p_h)$

Chan's Algorithm

But in general h is unknown!

ChanHull(P,m)

Divide P into sets P_i with $\leq m$ points

for i from 1 to $\lceil n/m \rceil$ **do** $\mathcal{O}(n/m)$

  Compute with GrahamScan $CH(P_i)$ $\mathcal{O}(m \log m)$

$\mathcal{O}(n \log m)$

$p_1 = (x_1, y_1) \leftarrow$ rightmost point in P

$p_0 \leftarrow (x_1, \infty)$

for $j = 1$ **to** m **do**

$\mathcal{O}(m) \cdot \mathcal{O}(n/m) = \mathcal{O}(n)$

  for $i = 1$ **to** $\lceil n/m \rceil$ **do**

$\mathcal{O}(\log m)$

    $q_i \leftarrow \arg\max\{\angle p_{j-1} p_j q \mid q \in P_i \setminus \{p_{j-1}, p_j\}\}$

  $p_{j+1} \leftarrow \arg\max\{\angle p_{j-1} p_j q \mid q \in \{q_1, \ldots, q_{\lceil n/m \rceil}\}\}$

$\mathcal{O}(n \log m)$

return $(p_1, \ldots, p_m)$

Total: $\mathcal{O}(n \log m)$

Chan's Algorithm

But in general h is unknown!

ChanHull(P,m)

Divide P into sets P_i with $\leq m$ points

for i from 1 to $\lceil n/m \rceil$ **do** $\mathcal{O}(n/m)$

  Compute with GrahamScan $CH(P_i)$ $\mathcal{O}(m \log m)$

$\mathcal{O}(n \log m)$

$p_1 = (x_1, y_1) \leftarrow$ rightmost point in P

$p_0 \leftarrow (x_1, \infty)$

for $j = 1$ **to** m **do**

$\mathcal{O}(m) \cdot \mathcal{O}(n/m) = \mathcal{O}(n)$

  for $i = 1$ **to** $\lceil n/m \rceil$ **do**

    $q_i \leftarrow \arg\max\{\angle p_{j-1} p_j q \mid q \in P_i \setminus \{p_{j-1}, p_j\}\}$ $\mathcal{O}(\log m)$

  $p_{j+1} \leftarrow \arg\max\{\angle p_{j-1} p_j q \mid q \in \{q_1, \ldots, q_{\lceil n/m \rceil}\}\}$

$\mathcal{O}(n \log m)$

return $(p_1, \ldots, p_m)$

Total: $\mathcal{O}(n \log m)$

Chan's Algorithm

But in general h is unknown!

ChanHull(P,m)

Divide P into sets P_i with $\leq m$ points

for i from 1 to $\lceil n/m \rceil$ **do** $\mathcal{O}(n/m)$

  Compute with GrahamScan $CH(P_i)$ $\mathcal{O}(m \log m)$

$\mathcal{O}(n \log m)$

$p_1 = (x_1, y_1) \leftarrow$ rightmost point in P

$p_0 \leftarrow (x_1, \infty)$

for $j = 1$ **to** m **do**

$\mathcal{O}(m) \cdot \mathcal{O}(n/m) = \mathcal{O}(n)$

  for $i = 1$ **to** $\lceil n/m \rceil$ **do**

$\mathcal{O}(\log m)$

    $q_i \leftarrow \arg\max\{\angle p_{j-1} p_j q \mid q \in P_i \setminus \{p_{j-1}, p_j\}\}$

  $p_{j+1} \leftarrow \arg\max\{\angle p_{j-1} p_j q \mid q \in \{q_1, \ldots, q_{\lceil n/m \rceil}\}\}$

  if $p_{j+1} = p_1$ **then return** $(p_1, \ldots, p_{j+1})$

$\mathcal{O}(n \log m)$

return failure

Total: $\mathcal{O}(n \log m)$

What to do with m?

Suggestions ...?

What to do with m?

FullChanHull(P)

for $t = 0, 1, 2, \ldots$ **do**

$m \leftarrow \min\{n, 2^{2^t}\}$

result $\leftarrow$ ChanHull(P, m)

if result $\neq$ failure **then** break

return result

What to do with m?

FullChanHull(P)

for $t = 0, 1, 2, \ldots$ **do**

  $m \leftarrow \min\{n, 2^{2^t}\}$

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Theorem. The convex hull $CH(P)$ of n points P in $\mathbb{R}^2$ can be computed in $O(n \log h)$ time with Chan's Algorithm, where $h = |CH(P)|$.

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What about the robustness of the algorithms?

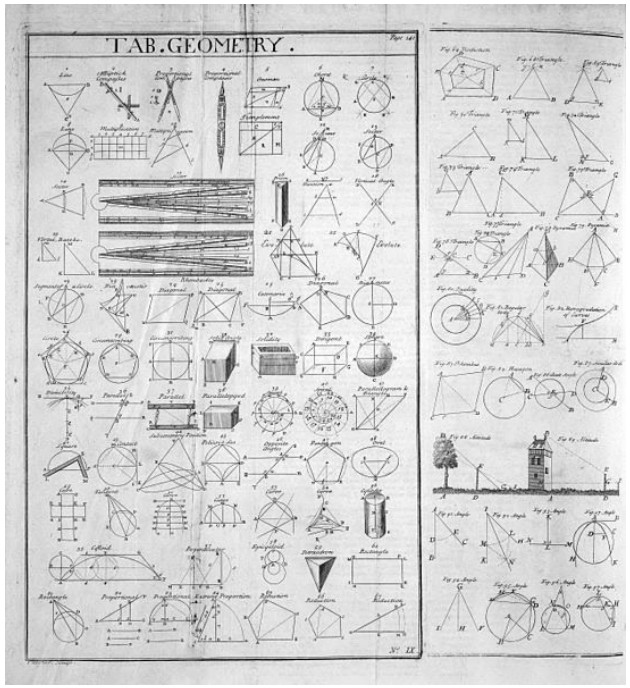
- Regarding robustness: imprecision of floating-point arithmetic
- FirstConvexHull possibly produces no valid polygon
- Graham and Jarvis always provide a valid polygon, but it may have minor defects

How to design Geometric Algorithm : Guidelines

1.) Eliminate degenerate cases ($\rightarrow$ *general position*)
- unique x (or y)-coordinates
- no three collinear points
- ...

2.) Adjust to degenerate inputs
- integrate into existing solutions
  (e.g., use lexicographic order if x-coordinates are not unique)
- may require special treament

3.) Implementation
- primitive operations (available in libraries?)
- robustness



Thank you!